

The Effects of focused deterrence on Crime and Community Outcomes in Philadelphia

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Abstract

focused deterrence is a crime-reduction strategy that combines a higher probability and severity of punishment with direct communication of consequences to likely offenders. These strategies have gained attention and wider implementation in recent years, driven by evidence of their impact on violent crime, but less is known about their effects on non-violent crime or their downstream effects on communities. We study Philadelphia's 2013 focused deterrence program, introduced in response to rising homicides, especially among young people, and estimate its effects on crime, rents, and housing values. Using a difference-in-differences design with propensity-score-matched census block groups, we find that the program reduced shooting victims by 41.3 percent and gun crime incidents by 24.2 percent. Total reported crime fell by 13.8 percent, reflecting declines in both violent and non-violent offenses. The policy also raised rents by 6.0 to 7.5 percent and housing values by 3.6 to 7.0 percent, consistent with improved safety being capitalized into local housing markets.

Keywords: Crime, Deterrence, Rental Markets, Housing Markets

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1 Introduction

Violent crime imposes substantial costs on both victims and the communities where it concentrates. Beyond the direct harms to victims, such as medical expenses, lost income, and criminal justice expenditures, violence shapes where households choose to live, where businesses choose to locate, and how neighborhoods develop over time. Traditional crime reduction methods, such as temporarily increasing the number of officers on the street, can be expensive and may not produce lasting reductions in crime ([Draca et al. \(2011\)](#); [Sorg et al. \(2013\)](#); [Sherman \(1990\)](#)). In response, a newer range of strategies has moved beyond traditional, broad-based policing toward more targeted approaches that aim to produce sustained reductions in crime, including hot-spot policing, environmental design interventions such as installing street lighting or restoring vacant lots, and community-led crime prevention efforts ([Braga et al. \(2019\)](#); [MacDonald et al. \(2026\)](#); [Nubani et al. \(2023\)](#)).

One such strategy, focused deterrence (FD), takes a distinct approach. Rather than concentrating enforcement on high-crime places, FD aims to change the behavior of a small number of high-risk offenders, such as gang members responsible for a disproportionate share of violence in an area. This is accomplished through a coordinated menu of law enforcement, community involvement, and social service actions, communicated directly to these offenders through call-in meetings and backed by swift enforcement when violence continues ([Braga et al. \(2018\)](#); [Braga and Weisburd \(2012\)](#); [Kennedy et al. \(1996\)](#)). While FD’s short-run effects on the violent crime it directly targets are well studied, less is known about its effects on other types of crime, such as non-violent offenses, whether its effects persist over longer horizons, and how the crime reductions it produces affect the communities where it is implemented.

Philadelphia adopted focused deterrence in April 2013 in response to rising violent crime in the city. Because FD’s effectiveness has varied considerably across the many cities that have adopted it ([Braga et al., 2018](#)), careful evaluations of individual implementations are important for understanding when and where the strategy works. This paper examines three interrelated questions about the effects of Philadelphia’s focused deterrence program. First, how effective is FD at reduc-

ing violent crime? Second, does FD impact other forms of crime such as non-violent crime? Third, to the extent that FD reduces crime, do these reductions capitalize into local housing and rental markets?

The targeted nature of FD in Philadelphia provides an appropriate context to causally estimate the impact this crime reduction strategy. To estimate the causal effect of Philadelphia’s focused deterrence strategy on crime and downstream neighborhood outcomes, we exploit geographic and temporal variation in the policy’s implementation across census block groups in Philadelphia. Because the program was implemented in a defined set of neighborhoods in Southern Philadelphia beginning in April 2013, we construct a matched comparison group using propensity score matching on pre-treatment demographics and crime trends from the pool of other Philadelphia County census block groups. Using this matched sample, we estimate event study and difference-in-differences specifications with block group and year fixed effects to estimate the dynamic and average treatment effects on treated block groups.

Using this strategy, we find that the FD policy had a significant impacts on crime and downstream outcomes in treated communities. In our main specification, we find the number of shooting victims declined by 41.35 percent and the total number of gun crime incidents fell by 24.15 percent. Total reported crime incidents declined by approximately 13.8 percent, which included reductions in both violent serious offenses (Part 1 offenses) and non-violent less-serious offenses (Part 2 offenses). We find these crime reductions produced by the FD strategy generated positive downstream effects on treated neighborhoods as reflected in local rental markets and residential property values. Following implementation of the strategy, rental prices increased by approximately 7.0 percent and housing values increased by approximately 6.0 percent. These estimates suggest the crime reduction attributable to the policy were capitalized into local property markets consistent with a broader literature linking changes in crime with changes to prices in rents and home values ([Thaler \(1978\)](#), [Linden and Rockoff \(2008\)](#), [Ihlanfeldt and Mayock \(2010\)](#), [Pope and Pope \(2012\)](#)).

Our findings contribute to the growing literature measuring the effectiveness of focused deterrence as a crime-reduction strategy. Meta-analyses by [Braga and Weisburd \(2012\)](#) and [Braga et al. \(2017\)](#) broadly find focused deterrence reduces violent crime across a wide array of implementations,

but most individual analyses focus on one outcome in the immediate post-treatment period, such as shooting victims or number of homicides. We contribute to this literature in three main ways. First, our setting allows us to causally identify the impacts of FD on crime, a feature that is missing from many analyses in the focused deterrence literature. Second, we expand the set of outcomes considered beyond the direct violent crime measures that exist in the current literature, allowing us to investigate whether FD generates spillovers onto non-violent offense types. Third, our setting allows us to examine the policy’s effects over a longer time horizon of six years, rather than the typical short-run two-year windows of prior evaluations, offering evidence of how focused deterrence’s crime-reduction effects evolve over time and whether crime reductions provide permeant changes in the community.

Our results on shooting victims and gun crime incidents align with this broader evidence of FD’s effectiveness at reducing violent crime, while our results on non-violent offenses suggest that focused deterrence generates spillover effects onto non-targeted offense types. Moreover, these effects persist over the long run, even after the official end of the policy, suggesting focused deterrence can have a lasting impact on crime in treated communities beyond the active intervention period.

More directly, our paper builds on the Philadelphia focused deterrence literature pioneered by [Roman et al. \(2019\)](#), extending their analysis with an updated panel, an alternative empirical framework, and downstream economic outcomes not previously examined in this setting. Our paper also speaks to the hedonic-capitalization literature on crime and housing prices ([Thaler, 1978](#); [Linden and Rockoff, 2008](#); [Pope and Pope, 2012](#)) by using a geographically targeted policy intervention to identify how crime reductions capitalize into housing values and rental prices.

The remainder of the paper proceeds as follows. Section 2 describes the Philadelphia focused deterrence intervention, the institutional details relevant for interpretation, and the timeline of call-ins and enforcements during the evaluation period. Section 3 reviews the literature on focused deterrence policies, and how crime impacts rental and housing markets. Section 4 describes the data sources, construction of the analytical sample, and the propensity-score matching process. Section 5 outlines the empirical strategy used in the main specification. Section 6 presents the main results of FD on crime and downstream outcomes. Section 7 analyzes the sensitivity of

our results to robustness checks such as alternative matching specifications. Section 8 discusses our results in terms of feasibility and magnitudes and measuring the benefits of the policy, while Section 9 concludes.

2 Policy Background

Focused deterrence policies are a type of crime reduction policy that aims to change offender behavior through a coordinated menu of law enforcement, community involvement, and social service actions (Braga et al. (2018); Braga and Weisburd (2012); Kennedy et al. (1996)). The strategy has been applied to three broad types of criminal behavior: violence committed by gangs and other criminally active groups, crime driven by street-level drug markets, and repeat offending by high-risk individuals (Braga et al. (2017), Kennedy et al. (1996), Kennedy (2012), Kennedy and Wong (2009), Deuchar (2013), Papachristos et al. (2007)). We focus on the first, group-involved violence, which is also the target of the Philadelphia program we study. These policies are constructed based on the theoretical framework of deterrence, which is premised on the assumption that criminals are rational actors and that crime can be deterred through a change in the costs or benefits of criminal activity (Becker (1968), Nagin (1998), Nagin (2013)). Violence-focused applications concentrate the threat of sanction on a small, identifiable population, typically gangs or criminally active street groups, responsible for a disproportionate share of violent activity in a given area, and advertise changes in the costs of violent crime to prevent offenses or re-offenses.

These policies are often implemented through a coordinated effort between different law enforcement agencies, legal agencies, city governments, local community figures, and research organizations. focused deterrence policies begin with interagency information sharing that identifies the hotspots of violent crime and the groups responsible for crime in those areas. Next, a menu of "levers" is bundled across different agencies to serve as a coordinated response to the continued violence committed by these groups. The policy formally begins with an initial call-in meeting, in which members of the targeted group are brought in to establish communication between these groups and law enforcement. This meeting then conveys that further violent crime will not be tolerated and

the new coordinated menu of punishments will fall to all members of the targeted groups in the event of further violent crime that is found in connection with the group. If further violent crime occurs after this call-in meeting, enforcement of the advertised menu of punishments is conducted swiftly to demonstrate credibility of the advertised consequences.

focused deterrence policies typically bundle five categories of levers ([Braga et al. \(2018\)](#)). Prosecutorial levers include higher bail requests, tougher plea bargains, and the use of federal prosecution to escalate the consequences of new offenses committed by targeted individuals. Police enforcement levers include intensified law enforcement activity in target areas, the execution of outstanding warrants against members of the targeted group following a triggering event, and the targeting of illegal sources of income associated with targeted individuals or groups. Probation and parole levers include increased supervision requirements for targeted individuals. Community-based levers include community moral voices, such as faith leaders, well-respected community groups, and family members of past victims of gun violence, who reinforce the anti-violence message. Finally, social service levers include voluntary education, vocational, and treatment programming for individuals who choose to disengage from violence. Across implementations, the distinguishing feature of focused deterrence is not the introduction of novel sanctions but the coordinated, swift, and credible deployment of legal authority that already existed across separate agencies, paired with community voices and the offer of legitimate alternatives through social services ([Kennedy et al. \(1996\)](#); [Roman et al. \(2019\)](#)).

The levers described above change the cost-benefit calculus of committing violent crime along multiple dimensions. On the cost side, three margins are affected: the probability of being caught, the severity of punishment, and the swiftness of punishment ([Nagin \(2013\)](#)). First, the perceived probability of being caught rises through several channels. Interagency intelligence sharing and intensified enforcement activity in target areas raise the likelihood that violent acts are detected and attributed, while direct communication to targeted groups signals that they are being monitored. Community involvement and awareness of the goal of reducing violence may also increase informal monitoring within neighborhoods. Group-level liability, in which any violent act attributable to a group triggers an enforcement response against all of its members, further raises the expected

cost of violence for each member. Second, the severity of punishment increases through the coordinated application of prosecutorial, police enforcement, and probation and parole levers to targeted individuals. Third, the swiftness of punishment increases through pre-coordinated enforcement protocols that allow agencies to respond to violent acts within days rather than weeks or months. In addition, social service offerings raise the opportunity cost of crime by improving returns to legitimate activity for targeted individuals.

Communication is the mechanism that activates these channels. In call-in meetings, targeted group members are required to attend and are explicitly informed of the enforcement consequences awaiting them and their group. These meetings are designed to update offenders' subjective beliefs about the probability, severity, and swiftness of sanction. The message is then intended to diffuse through group networks, extending this updating to members who did not attend and to potential offenders in the broader community.

Following substantial increases in violent crime in 2006 and again in 2010, the Mayor of Philadelphia, in conjunction with the Philadelphia Police Department (PPD) and the Philadelphia District Attorney's Office (DAO), began reviewing potential violence reduction strategies. Beginning in the fall of 2012, this working group became familiar with the focused deterrence strategy developed during Operation Ceasefire and began shaping an analogous intervention for Philadelphia. The working group expanded to include the Adult Probation and Parole Department, Juvenile Probation, the Philadelphia Housing Authority Police Department, social service agencies, researchers from Temple University, and the local federal prosecutor's office ([Roman et al. \(2017\)](#); [Roman et al. \(2019\)](#)).

Philadelphia's intervention was specifically targeted at the area of South Philadelphia. South Philadelphia was not inherently the area with the highest amount of homicides in the city, and instead was selected based of several factors. The working group's selection of South Philadelphia reflected several factors: an entrenched street gang culture, pre-existing interagency relationships among local law enforcement, manageable geographic coverage, clustered gang territories, and a pre-existing structured intelligence apparatus within PPD's Central Intelligence Unit ([Roman et al. \(2017\)](#); [Roman et al. \(2019\)](#)).

The intervention officially launched with an initial call-in meeting on April 17, 2013, at which time 16 active gangs became the focus of the program; the first enforcement was conducted three days later in response to a shooting committed by a targeted group ([Roman et al. \(2019\)](#)). Over the first two years of the policy, four call-in meetings and 16 enforcements were carried out, with ten gangs subject to enforcement actions, some on more than one occasion. The program remained active until early 2017.

Philadelphia’s implementation deployed each of the general lever categories described above, prosecutorial, police enforcement, probation and parole tools, community involvement, and social service provisions, while also introducing unique levers specific to this implementation of FD. First, housing and utilities-related sanctions included termination of public utility services for nonpayment or illegal connections, review of public housing eligibility, and targeted code enforcement. Second, probation and parole escalation tools included increased supervision intensity, execution of outstanding warrants, revocation of probation for those arrested but not yet convicted of a new offense, and enforcement of outstanding child support orders. Third, prosecutorial tools included requests for high bail following any new arrest and advocacy for longer or more restrictive sentences upon new convictions ([Roman et al. \(2019\)](#)). Social services complemented these levers. [Roman et al. \(2019\)](#) report that services made initial contact with approximately 112 group members across 14 street groups between April 2013 and March 2015, with roughly 33 percent engaging in some level of services such as GED programs, substance abuse treatment, or vocational training.

3 Literature Review

focused deterrence originated in Boston in the mid-1990s as Operation Ceasefire, a coordinated effort to reduce youth gun violence that paired a deterrence strategy with direct law enforcement action against illicit firearm trafficking ([Kennedy et al. \(1996\)](#)). Boston’s initial pilot was followed by a 63 percent decline in mean monthly youth homicides ([Braga et al. \(2017\)](#)). Since then, the strategy has been implemented in 84 U.S. cities, ranging from large metropolitan areas such as Chicago and Los Angeles to smaller cities such as New Haven, Connecticut and Roanoke, Virginia

(Braga et al. (2018), Papachristos et al. (2007), Tita et al. (2011), Sierra-Arevalo et al. (2017), Saunders et al. (2015)). This proliferation has generated a substantial body of evidence with which to evaluate the strategy’s effectiveness. Meta-analyses of this literature find that focused deterrence strategies are associated with a statistically significant, moderate overall crime reduction effect. The strongest impacts are observed in programs targeting serious violence by gangs and criminally active groups, while strategies aimed at repeat offenders more broadly or at drug market crime tend to produce more modest reductions (Braga and Weisburd (2012); Braga et al. (2018)).

The existing evaluation literature on Philadelphia’s focused deterrence intervention originates with Roman et al. (2019), who assess the community- and gang-level impacts of the intervention using a propensity score matching design at the census block group level. They match treated block groups in South Philadelphia to untreated block groups elsewhere in the city on pre-intervention (2012) level of shootings, police pedestrian and cars tops, number of street gangs, and demographic characteristics of block groups. They find a statistically significant 35 percent reduction in shooting rates in treated areas over the 24 months following the April 2013 call-in. The gang-level findings, however, are more mixed: while some models show significant reductions in shootings around the territories of targeted gangs, the majority of specifications do not yield statistically significant gang-level effects, suggesting that the community-level deterrence effect may operate through general deterrence mechanisms rather than through behavioral change among specifically targeted groups

Subsequent work has examined the mechanisms behind these crime reductions. Because the strategy is intended to rely on communication and the threat of sanctions rather than on high arrest volumes, a key concern is whether enforcement in treated areas was overly aggressive. Roman et al. (2020) address this by providing a descriptive account of the enforcement levers used in Philadelphia, comparing arrests around targeted group territories after four enforcement actions to arrests around matched comparison groups. They find no statistically significant differences in overall arrests or arrests by offense type and conclude that enforcement was not overly aggressive. Other work has examined how responses to the deterrence message varied across targeted groups, finding that group-level social media activity partly explains the variation (Hyatt et al. (2021)), and has documented the spatiotemporal patterns of shooting hot spots before, during, and after

implementation (Boschan and Roman (2024)). Taken together, this literature primarily measures the impacts of focused deterrence in terms of crime and enforcement outcomes. Much less is known about what reductions in violence mean for the communities where these programs operate, including whether improved neighborhood safety is capitalized into local housing markets.

A long literature in urban and public economics treats crime as a localized disamenity that is capitalized into housing markets. Thaler (1978) provides early evidence that households pay for crime control, and later studies use designs that more directly address the endogeneity of crime and neighborhood change. Gibbons (2004) finds roughly a 10 percent decline in property values in London for a one-standard-deviation increase in local property crime. In the U.S., Ihlanfeldt and Mayock (2010) use a nine-year Miami-Dade panel with first-difference IV methods and find that robbery and aggravated assault, rather than all crime categories, materially depress housing values, with elasticities of roughly 0.1 to 0.3. Work exploiting the 1990s crime decline estimates elasticities of property values between about 0.15 and 0.35 (Pope and Pope (2012)). Localized penalties also appear around sex offender registry disclosures (Linden and Rockoff (2008), Pope (2008)). This literature suggests that public safety is priced into local housing demand, with the clearest evidence in sale prices and housing values and thinner direct evidence on rents.

This paper contributes to both strands of literature. Relative to existing evaluations of Philadelphia’s focused deterrence program, we match treated and control block groups on pre-intervention trends rather than on a single pre-intervention year of levels, and we extend the evaluation beyond shooting victims to violent and non-violent crime incidents, rents, and housing values. We also extend the post-policy period through 2019, well beyond the two years previously examined, and assess the sensitivity of our estimates to the choice of control group using alternative matching and weighting strategies. Relative to the literature on crime and housing markets, we study the effects of a place-based policing intervention on both housing values and rents, where direct evidence is thinner. Because prior work documents reductions in violence at the community level, neighborhood-level economic outcomes are a natural next margin to study for focused deterrence policies.

4 Data

4.1 Data Sources

To analyze the impact of Philadelphia’s focused deterrence Intervention on crime and downstream effects on treated communities, we construct a census block group by year panel of Philadelphia County spanning from 2006 to 2019. The census block groups in our data are defined using the 2010 census block group definitions with crosswalks for pre-2010 block groups completed using IPUMS’s National Historical Geographic Information System (NHGIS) block group crosswalks ([Schroeder et al. \(2025\)](#)). Treatment assignment data at the block group level were collected from [Roman et al. \(2017\)](#) with the policy beginning in April 2013 and ending in late 2017. We end our sample in 2019 to avoid confounding issues from the COVID-19 pandemic, which produced well-documented disruptions in both crime patterns and policing behavior ([Regalado et al. \(2022\)](#)).

Crime is measured through two different categories of outcomes: the number of crime incidents and the number of shooting victims. A crime incident is a reported offense made to the police in which the offense is committed by the same offender, or group of offenders acting in concert, at the same time and place ([Federal Bureau of Investigation \(2019b\)](#)). Shooting victim data are recorded separately, with one observation per victim, as a single shooting incident may involve multiple victims. Crime incident and shooting victim data are collected from the Philadelphia Police Department’s (PPD) Crime Incidents and Shooting Victims datasets([Philadelphia Police Department \(2016a\)](#); [Philadelphia Police Department \(2016b\)](#)). Crime incident data are available for all of Philadelphia County at an incident-by-address level over the full sample period (2006-2019), while shooting victim data are available at the victim-by-address level beginning in 2010. These data are then aggregated by crime incident and shootings categories, census block group, and year level.

To examine the downstream effects of the policy on community-level outcomes, we gather contract rent and housing value data from the American Community survey (ACS) 5-year estimates at the census block group level from 2009 to 2019 ([U.S. Census Bureau \(2019\)](#)). We acknowledge that the year-to-year variation in ACS-based outcomes may be attenuated given the rolling nature of

the five-year estimates; however, this dataset allows us to view these crucial outcomes at a granular level.

We draw on the same ACS 5-year estimates to construct demographic and socioeconomic information at the census block group level, including measures of population, income, educational attainment, and racial composition, which we use both as matching covariates in the propensity score matching procedure, and as controls in estimation of the policy’s treatment effect.

We have several different outcomes of interest. First, we investigate the number of shooting victims within a census block group within each year. Second, we investigate how focused deterrence impacted the total number of crimes in a census block group each year. We then break down these crimes by category based on the Federal Bureau of Investigation (FBI) Uniform Crime Reporting categories ([Federal Bureau of Investigation \(2019b\)](#)). These categories include part one offenses, which include serious crimes against people and property — homicide, rape, robbery, aggravated assault, and arson — and part two offenses, which include other crimes against persons (assaults of a non-aggravated nature), public disorder crimes (vandalism, prostitution, drunkenness, etc.), drug and alcohol offenses, white collar crimes (fraud, embezzlement, forgery), and all other crimes (weapons violations, possessing stolen property, state or local laws not otherwise specifically identified).¹ We also construct a crime measure, the number of crimes involving a firearm, which is the total number of aggravated assaults with a firearm, robberies with a firearm, weapons violations, and homicides. The crime incident dataset does not specify the weapon used for homicides; however, the FBI reports that around 70 percent of homicides are committed with a firearm ([Federal Bureau of Investigation \(2019a\)](#)). Regardless, this measure might overstate the count of gun crime incidents. As a robustness check, we adjust the definition of gun crimes in several ways.

Third, we investigate the downstream effects of crime reduction on rental and housing markets in treated communities. Our primary outcomes of interest are contract rent and housing structure value, both measured in 2019 real dollars. For each outcome, we examine impacts at multiple points of the distribution — lower quartile, median, and upper quartile. These distributions are within a census block group, not across a broader geographic range such as the county. We use these different

¹A full listing of FBI UCR offense categories and their definitions is available in [Federal Bureau of Investigation \(2019b\)](#).

points of this distribution to explore heterogeneous effects of crime reduction in these communities.

4.2 Summary Statistics

Figure 1 shows the number of criminal homicide incidents by census block group for Philadelphia county in 2011, 2-years prior to the implementation of the focused deterrence intervention. As mentioned in the policy background, Southern Philadelphia was chosen for intervention based on entrenched street gang culture, pre-existing interagency relationships among local law enforcement, manageable geographic coverage, and clustered gang territories (Roman et al. (2017); Roman et al. (2019)). As seen by the figure, several census block groups in Southern Philadelphia are "hot spots" for criminal homicides as these several communities experienced between 1 to 6 homicides in 2011. However, homicides in Philadelphia county are not located exclusively in Southern Philadelphia in frequency or severity as both Western Philadelphia and Northern Philadelphia have census block groups that experienced homicides, and census block groups that experienced large severity in the number of homicides. This geographic dispersion of homicides and shootings around Philadelphia helps motivate our propensity score matching technique, as we will be able to match similar census block groups in Southern Philadelphia to other census block groups around Philadelphia in terms of crime incident, shooting victim, and demographic trends.

Table 1 compares the pre-treatment means of shootings victims, crime incidents, and community characteristics of South Philadelphia to the rest of the city. Compared to the rest of Philadelphia, Southern Philadelphia has statistically fewer shooting victims, Part 1 (violent) crime incidents, Part 2 (non-violent) crime incidents, and gun crime incidents. In addition, Southern Philadelphia census blocks have lower Black and Hispanic population shares, a higher share of the population with a Bachelor's degree or above. These census blocks also have a higher labor force participation rate, lower unemployment rate, smaller shares of households on SNAP benefits and below the poverty line, and higher median household income. These differences in shooting victims, crime incidents, and community characteristics once again highlights the necessity of identifying a comparable comparison group to Southern Philadelphia, as these communities look noticeably different in observables on average from the rest of the city.

Table 2 compares pre-treatment means of contract rents and housing values in South Philadelphia to the rest of the city. Lower quartile rents are similar across the two areas, but the distributions diverge above that point: median and upper quartile rents in South Philadelphia exceed those in the rest of the city by approximately \$25 and \$80, respectively. Housing values are significantly higher across the entire distribution, with lower quartile, median, and upper quartile values in South Philadelphia exceeding the rest of the city by roughly \$37,000, \$ 64,000, and \$ 94,000, respectively. As with the demographic and crime comparisons above, these level differences in housing and rental markets motivate our propensity score matching approach over naive comparisons to the rest of the city.

In addition to the pre-treatment averages of these characteristics being noticeably different, the trends in the shootings and crime incidents during the pre-treatment years are also different. Figure 2a shows the trend in the average number of shooting victims in Southern Philadelphia compared to the rest of Philadelphia. While both areas were experiencing around an average of 1.2 shooting victims in a census block in 2010, Southern Philadelphia was already experiencing a noticeable decline in the average number of shooting victims leading up to the intervention in 2013 that was not seen in the rest of the city.

Figure 3 and Figure 2b show the trends in crime incidents in Southern Philadelphia compared to the rest of the city broken down by total, Part 1, Part 2, and gun crime incidents. Once again there are noticeable differences in the trends of crime incidents during the pre-treatment period. While the trends in Part 1 crime incidents look comparable across the two groups, Southern Philadelphia was experiencing a downward trend in the average number of total crime incidents, Part 2 crime incidents, and gun crime incidents not seen in the rest of the city as early as 2010. Therefore, a naive comparison utilizing all other Philadelphia census block groups to Southern Philadelphia in a causal analysis would struggle to pass the parallel trends assumption necessary to calculate an accurate treatment effect of focused deterrence.

Figures 4 and 5 show trends in contract rents and housing values in South Philadelphia and the rest of the city, broken down by lower quartile, median, and upper quartile. In contrast to the crime trends, which diverge between the two areas, pre-treatment rental prices and housing values appear

visually comparable. Both series show a sharp decline from 2009 to 2010 reflecting the post-Great Recession housing market correction, followed by a slow recovery. Beginning around 2013, South Philadelphia experiences sharper growth in rents across the distribution; a similar divergence in housing values emerges around 2015. While this post-treatment divergence is suggestive, attributing it to focused deterrence requires isolating a control group with comparable pre-treatment crime trends, which our matching procedure constructs.

4.3 Propensity Score Matching

To begin the process of finding a causal treatment effect of focused deterrence on the number of shooting victims, the number of crime incidents, and downstream effects, we utilize propensity score matching. Specifically, we implement a nearest-neighbor matching technique with replacement which identifies census block groups among the potential pool of communities from the rest of the city based on the estimated probability of treatment selection. We estimate the probability of treatment selection with the following equation:

$$Pr(Treat_i) = \Lambda(\alpha + \beta_1 \Delta Part1_i + \beta_2 \Delta Part2_i + \beta_3 \Delta Gun_i + \gamma \bar{X}_i + \epsilon_i) \quad (1)$$

where $Pr(Treat)$ is the probability of census block group i ever receives treatment. This probability was estimated using a logistic regression as denoted by Λ . The covariates used to predict the probability of treatment were selected based on justification of treatment selection as outlined in [Roman et al. \(2019\)](#). First, $\Delta Part1_i$, $\Delta Part2_i$, and ΔGun_i represent the trend (first difference) in Part 1, Part 2, and gun crime incidents from 2006 to 2012 in census block group i . Second, $\bar{X}$ represent a vector of community characteristics including youth male density (share of population that is male between the ages of 14 and 24), total population, Black population share, Hispanic population share, and economic disadvantage index (created using a census block group’s unemployment rate, share of households below the poverty line, and share of households on SNAP benefits). These community characteristics used as covariates in this matching process are averages of these values between the years of 2010 to 2012. The years utilized in the creation of the crime trends,

and community characteristic averages were selected based on of data availability in from PPD and ACS. Finally, census block groups located within the 22nd Police District and several block groups within the 30th Police District were not allowed to be matched on due to their exposure to Philadelphia CeaseFire, a coinciding shooting reduction policy happening in a small area in central Philadelphia (Roman et al. (2018)).

Table 3 and Figure 6 show the comparison between the unmatched and matched samples as a result of the above matching procedure. The unmatched sample shows substantial and statistically significant imbalance across nearly every covariate, particularly in Part 2 crime trends, percent Black, and percent Hispanic. Southern Philadelphia had steeper declining crime trends, were less Black and less Hispanic compared to the broader pool of the rest of Philadelphia. From the 1,182 possible comparison block groups, the matching process found 119 comparable census block groups and substantially reduced bias across almost every covariate. The one caveat is that after matching, percent Black still has a statistical difference in means between the treated and comparison group. In the matched sample, census block groups in Southern Philadelphia statistically have a higher black population share compared to the match comparison group.

Propensity score matching techniques rely on two assumptions - conditional independence and common support. While conditional independence is not inherently testable, we know that South Philadelphia was selected for treatment based on the observable characteristics of entrenched street gang culture, pre-existing interagency relationships among local law enforcement, manageable geographic coverage, and clustered gang territories (Roman et al. (2017); Roman et al. (2019)). Covariate selection was based on capturing these observable determinants of selection. Common support is directly testable. Figure ?? shows the kernel density of propensity scores for the entire sample of census block groups in Philadelphia. Census block groups in South Philadelphia and the rest of the city have possible range of propensity scores ranging from 0 to 1, providing evidence of common support. Of note, there is some bunching of propensity scores in the rest of Philadelphia that is near zero, speaking to the fact that many census block groups in Philadelphia are not similar to the treated units in Southern Philadelphia. Figure ?? shows another kernel density of propensity scores in the post-matched sample. There is near perfect overlap of propensity scores among both

the treated and control samples across the possible ranges of propensity scores from 0 to 0.85.

Visually, the resulting treated and control groups can be seen on the map of Philadelphia in Figure 8. The matched control group selected census block groups from different areas around Philadelphia predominantly in western, northwestern, and northeastern Philadelphia. Only a few census block groups adjacent to the Southern Philadelphia treatment border were selected as control groups, alleviating some concerns of direct crime spillover effects of the intervention into control block groups. For robustness, we remove these control bordering block groups.

Figures 9 and 10 show the timeseries of the average shooting victims, and crime incidents among the treated and control groups in the matched sample. Compared to the pre-matched sample the trend in average number of shooting victims per year is similar across both groups prior to the intervention. Following the start of the policy in 2013, there is a noticeable decline in the average number of shooting victims for the first two years of the intervention not seen in the control group. After 2014, the treated areas experience a rise in average shooting victims and then follow a similar trend to control group for the rest of the sample.

In terms of average crime incidents, we now witness a more comparable pre-policy trend in average total number of crime incidents, Part 1, Part 2, and gun crime incidents in the matched sample. The average total number of crime incidents experiences a drop in the years following the implementation of the FD intervention not seen in the control group, with this drop being primarily driven by Part 2 crime incidents. While Part 1 crime incidents follow similar trends post-implementation, we witness a spike in Part 1 Crime incidents in the control group during 2016 in the control group that is not seen in the treated group which might speak to the policy having an effect of maintaining a downward trend in particularly violent crimes during a period of growing violence in the control group. The trend in the average gun crime incidents declines in the first two years of the policy in the treated block groups compared to the control block groups (Figure 9b). Following 2014, the average number of gun crime incidents trend in a similar fashion among both groups for the rest of the period.

Table 4 presents the raw means of the outcomes of interest for the control and treated group in the pre-intervention period and post-intervention period. Comparing the change in the average

number of shooting victims by group, the control group experienced an increase of shooting victims from an average of 0.179 shooting victims in a census block group per year, an increase of 164.24 percent. Meanwhile, the treated group demonstrated an increase in the average number of shooting victims from 0.387 to 0.592, an increase of 52.97 percent. Although the number of shooting victims increased for both groups, there is descriptive evidence that South Philadelphia experienced a slowed growth in number of shooting victims. In addition, in terms of crime incidents both groups experienced declines in the average total number of crime incidents and within each category of crime. South Philadelphia experienced larger reductions in crime with a 31.67 percent decline in average total crime incidents, 22.25 percent decline in Part 1 crime incidents, 36.94 percent decline in Part 2 crime incidents, and 41.37 percent decline in the average number of gun crime incidents. Meanwhile, the control group experienced a 15.95 percent decline in average total number crime incidents, 16.05 percent decline in Part 1 crime incidents, 15.86 percent decline in Part 2 crime incidents, and 20.62 percent decline in gun crime incidents respectively. This raw means comparison provides partial descriptive evidence of the policy’s effectiveness.

Table 5 presents the post-matching raw means of the rental and housing value outcomes for the control and treated groups in the pre-intervention and post-intervention periods. Beginning with rents, the control group experienced modest growth across the distribution: lower quartile rents rose from \$726.01 to \$771.45, an increase of 6.26 percent; median rents rose from \$913.24 to \$957.06, an increase of 4.80 percent; and upper quartile rents rose from \$1,123.30 to \$1,186.40, an increase of 5.62 percent over this period. The treated group experienced substantially larger growth across the same points of the distribution, with lower quartile rents rising from \$651.93 to \$740.14 (13.53 percent), median rents rising from \$835.71 to \$948.56 (13.50 percent), and upper quartile rents rising from \$1,065.60 to \$1,208.16 (13.38 percent).

Turning to housing values, both groups experienced declines at the median and upper quartile, consistent with the broader housing market correction; however, the treated group experienced smaller declines, and at the lower quartile experienced growth. In the control group, lower quartile housing values declined from \$176,051 to \$173,553 (1.42 percent), median housing values declined from \$250,280 to \$228,649 (8.64 percent), and upper quartile housing values declined from \$336,164

to \$311,539 (7.32 percent). In the treated group, lower quartile housing values rose from \$169,537 to \$177,459 (4.67 percent), median housing values declined only modestly from \$246,398 to \$242,500 (1.58 percent), and upper quartile housing values declined from \$338,636 to \$326,019 (3.73 percent).

Across all six outcomes, the treated group experienced more growth than control group in the post-intervention period. The treatment-control gap in rent growth is approximately uniform across the distribution, at roughly 7 to 9 percentage points. The treatment-control gap in housing values is more concentrated in the lower and middle of the distribution—approximately 6 percentage points at the lower quartile and 7 percentage points at the median, compared to roughly 4 percentage points at the upper quartile. In the next section, we outline our empirical strategy to determine causally how much of these differential trends in crime and downstream effects can be attributed to the focused deterrence intervention.

5 Empirical Strategy

5.1 Event Study

To investigate the dynamic treatment effects and pre-treatment trends of the focused deterrence policy, we estimate the following equation:

$$y_{i,t} = \sum_k \beta_k D_{i,t}^k + \alpha_i + \delta_t + \lambda X_{i,t} + \epsilon_{i,t} \quad (2)$$

where $y_{i,t}$ is either the number of shooting victims, number of crime incidents, or downstream effect outcome of interest in census block group i in year t . $D_{i,t}^k$ is an indicator of whether census block group i is in its k -th year relative to treatment at time t . The reference year is one year prior to the policy, 2012. α_i and δ_t represent census block group and year fixed effects respectively. Finally, $X_{i,t}$ is a vector of controls including demographics. Standard errors are clustered at the census block group level.

5.2 Difference-in-Differences

To investigate the static average treatment on treated (ATT) effect of the focused deterrence policy, we estimate the following equation:

$$y_{i,t} = \beta(Post \times Treat)_{i,t} + \alpha_i + \delta_t + \lambda X_{i,t} + \epsilon_{i,t} \quad (3)$$

where $y_{i,t}$ is either the number of shooting victims, number of crime incidents, or downstream effect outcome of interest in census block group i in year t . $(Post_t \times Treat_i)$ is an interaction indicator of $Post$ which is equal to 1 if the year is equal to or after 2013, and $Treat$ is equal to 1 if the census block group ever received the focused deterrence treatment. α_i and δ_t represent census block group and year fixed effects respectively. Finally, $X_{i,t}$ is a vector of controls including demographics. Standard errors are clustered at the census block group level.

In this specification, β represents the average treatment effect on the treated (ATT) of focused deterrence on the shooting, crime, or downstream outcome of interest. Identification of this effect relies on four assumptions: parallel trends between treated and control block groups, the stable unit treatment value assumption (SUTVA), no pre-treatment anticipation, and composition stability.

The parallel trends assumption is concerned with how treated block groups would have evolved after April 2013 in the absence of the policy. This assumption cannot be tested directly; however, we have three pieces of supporting evidence that this assumption holds in this setting. First, propensity score matching on pre-intervention crime trends and demographics yields treated and control block groups that are comparable on these dimensions. Matching cannot account for unobserved characteristics, but pre-period trends in shooting victims and crime incidents are likely to capture much of what unobserved factors imply for local violence, so we view the matched observables as good approximations for the underlying risk of gun violence in these areas. Second, raw means before and after matching show that matching substantially improves the comparability of pre-policy crime trends. Third, event-study estimates allow us to examine pre-treatment differences between the treated and control groups as seen in the next section. Because we match on pre-period trends, flat pre-treatment event-study coefficients are partly mechanical, so we interpret them as suggestive

evidence and not as proof of parallel trends.

SUTVA requires that outcomes in control block groups are unaffected by treatment in South Philadelphia. Two channels could violate this assumption, and they bias our estimates in opposite directions. Displacement of crime from treated to control areas would worsen control outcomes and overstate the policy’s effects, whereas diffusion of deterrence or capitalization of safety improvements into nearby control areas would improve control outcomes and understate them. Several features of the setting limit both concerns. Shootings attributed to group violence are highly localized around neighboring group territories, and these territories tend to be stable over time (Roman et al. (2017)), which makes relocation of the violence targeted by the policy unlikely. Our descriptive raw means also do not display any jumps in crime in the control group following the policy, which is suggestive that crime did not simply relocate to other parts of Philadelphia due to the implementation of the policy. In addition, South Philadelphia is separated from Northern Philadelphia by Interstate 676 and from Western Philadelphia by the Schuylkill River (Figure 8), which limits direct geographic spillovers. For robustness, we analyze our results adjusting our comparison group by removing any adjacent census block groups to Southern Philadelphia. SUTVA also requires a well-defined treatment. Because enforcement intensity varied across targeted groups, our estimates should be interpreted as average effects across the treated area and not as the effect of a single uniform treatment.

As described in the policy background, the working group settled on the focused deterrence model and assembled the required interagency coordination in the fall of 2012, and planning continued into early 2013 (Roman et al. (2017, 2019)). This planning was conducted among agency partners, and the first message to targeted groups came with the initial call-in meeting in April 2013, with the first enforcement following shortly after. This timeline makes anticipatory behavioral changes among offenders, particularly in group-involved crime, unlikely before April 2013. The event-study coefficients for the periods immediately preceding the call-in provide an empirical check on this assumption.

Identification also requires that the composition of the population and housing stock in treated and control block groups does not change differentially as a result of the policy. Treated and control

block groups are otherwise stable over the sample: once a block group is treated in 2013, it remains treated to the end of the sample. Control block groups are not exposed to focused deterrence, and the matching specification has already accounted for census block groups that were exposed to concurrent crime-reduction policies elsewhere in the city. Citywide policies are absorbed by time fixed effects. With these assumptions in mind, the next section presents results from both specifications (event study and difference-in-differences results) grouped by outcome.

6 Results

6.1 Shootings and Crime Results

First, we present the results of the event study specification on the number of shooting victims and crime incidents. Figure 13a presents the event study results for the number of shooting victims from 2010 to 2019. Prior to the implementation of the FD policy in 2013, the trends between the treated and control groups are comparable. During the first two years of the intervention there is a decline in the average number of shooting victims by about 0.4 to 0.5 victims per year in the treated census block groups compared to the control groups; however, these estimates are noisy and not statistically significant. Following 2014, the treated group starts to trend back towards zero with the point estimate never returning to its pre-treatment difference. The policy officially ended in 2017, and the differences in trends of shootings appear to persist for two years following the end of the policy.

Figure 14 and Figure 13b show the event study results for the number of crime incidents from 2006 to 2019. For most of the crime incident categories — total, Part 1, and Part 2 — the trends between South Philadelphia and the matched control group are similar. The trend in gun crime incidents appears to be parallel between the two groups from 2009 onwards, but prior to that the trend in gun crime incidents was higher in Southern Philadelphia compared to the matched control group. Following the implementation of the FD intervention, there is a noticeable drop in the number of total crime incidents in around 20 crime incidents per year on average in the treated census block group compared to the control. This drop in total crime incidents is attributed to drops

in Part 2 Crime incidents, while Part 1 Crime incidents show potential increases in crime incidents but are not statistically significant for any year. In addition, although Part 1 Crime incidents in aggregate don't appear to be impacted by the policy, the number of gun crime incidents does show a statistically significant decline post-implementation of the policy from 2014 to 2019.

Table 6 shows the results of the two-way fixed effects difference-in-differences specification. For this specification, we calculate the average treatment effect on the treated for shooting victims from 2010 to 2017, and for crime incidents from 2006 to 2017. This is to capture the strict treatment effect during the policy's duration, and not any reversion of these trends after the policy was discontinued. Starting with the average number of shooting victims per census block group per year, we find that the FD intervention is attributed to a decline of 0.373 shooting victims compared to the control group. Compared to the pre-treatment period mean, this is a reduction of 41.35 percent.

For crime incidents, the FD intervention is associated with a decline of 19.165 total crime incidents on average in treated census block groups compared to control groups. This reduction represents a 13.84 percent decline in total number of crime incidents compared to the pre-treatment mean. Breaking this reduction down by crime incident type, Part 1 crime incidents increased by 0.958 incidents on average per year in treated census block groups compared to the control group; however, this increase is not statistically significant. On the other hand, Part 2 crime incidents demonstrated a 20.123 decrease on average in treated census block groups compared to control block groups. This reduction represents a 22.66 percent decline relative to the pre-treatment Part 2 crime incident mean.

Focusing on gun crime incidents specifically, the focused deterrence policy is associated with a decline of 1.298 gun crime incidents and is statistically significant at the 1 percent level. Compared to the pre-treatment mean, this decline represents a 24.16 percent reduction in the number of gun crime incidents in treated census block groups compared to control block groups. These results combined with the reduction in shooting victims would suggest that the intervention had a noticeable impact on reducing shooting activity in treated communities.

Tables 7 and 8 break down Part 1 and Part 2 crime categories into further subcategories. Part 1 crimes are defined as serious offenses against persons or property and are broken down into violent

crimes (serious crimes against persons) and property crimes (serious crimes against property). While we see a slightly positive, statistically insignificant effect on Part 1 crimes overall, violent crimes experience a statistically significant decline of 1.859 incidents on average per year, or around 14.2 percent relative to the pre-treatment mean, in treated census block groups compared to the matched control block groups. Property crimes, on the other hand, experienced a statistically insignificant and imprecise increase over the same period. Overall, this suggests that the FD policy had a significant impact on reducing violent crimes in treated census block groups but had no measurable impact on property crime incidents.

Part 2 crimes are generally less serious or nonviolent offenses and have been broken down into the following subcategories: other crimes against persons, public disorder crimes, drug and alcohol violations, white collar crimes, and a catchall other crimes subcategory. The overall decline in Part 2 crimes is primarily driven by large reductions in public disorder crimes (19.2 percent), drug and alcohol crimes (31.9 percent), and other crimes (32.8 percent). The reductions in Part 2 crime incidents, which drive a large portion of the decline in total crime incidents, suggest that the intervention contributed not only to reductions in violent and shooting crimes, but also deterred a number of nonviolent offenses.

6.2 Downstream Effects

Next, we present the results of the downstream effects of focused deterrence as measured through rental prices and housing values via ACS 5-year estimates. As a reminder, both the rental prices and housing valuations represent 5-year rolling average values and are subject to attenuation bias. We anticipate the magnitude of these effects to grow over time as the share of post-policy years in each rolling window increases, and as long-run mechanisms, such as rental price stickiness and gradual information updating about improved neighborhood safety among renters and buyers, fully materialize. Therefore, we consider these effects to be a lower bound for the potential effects of focused deterrence on these downstream outcomes.

First, Figure 15 presents the event study specification results for lower quartile rents, median rents, and upper quartile rents. For lower and upper quartile rents, the rental price trends are

comparable, as both groups experienced similar declines following the recession in 2009, 2010, and 2011. Following the implementation of the focused deterrence policy, both lower and upper quartile rents grew in the treated census block groups compared to the control block groups. Both of these groups experienced growth one year after the policy’s implementation, in 2014, and then became statistically different from the control group from 2015 until the end of the sample. The trend in median rental prices between the two groups is not comparable in the pre-treatment period in 2011, but follows a similar rental price growth trajectory to the other parts of the rental distribution following the implementation of the policy in 2013.

Figure 16 presents the event study specification results for lower quartile, median, and upper quartile housing valuations. Across all three outcomes, the trend in housing valuations is comparable between the treated and control group. Following the implementation of the focused deterrence policy in 2013, we see growth among lower quartile and median housing valuations. Lower quartile and median housing valuations increased starting two years after the implementation of the policy in 2015, and the difference generally continued to grow through the end of the sample. Lower quartile housing valuations experienced a sharper increase following the implementation of the policy compared to the two other areas of the distribution. Upper quartile housing valuations appear to have no statistically significant difference between the treatment and control group in the immediate years following the implementation of the focused deterrence policy. Upper quartile housing valuations between the two groups began to diverge at the end of the sample in 2018 and 2019 following the end of the policy.

Table 9 shows the results of the two-way fixed effects difference-in-differences specification for the downstream effects on rental prices and housing valuations. Unlike for the crime and shooting results, we calculate the average treatment effect on rental prices and housing valuations using the 2009–2019 sample. This is done to capture the full treatment effect of the policy adjusting for any delayed downstream effects due to potential stickiness in the rental/housing market, or for the diffusion of information about declining crime in the census block groups.

The first three columns of Table 9 show the average treatment effect on the treated of the policy on rental prices across the lower quartile, median, and upper quartile distribution areas. We find

statistically significant increases in rental prices following the implementation of the FD policy of \$41.60 among lower quartile rental price units, \$64.31 among median rental price units, and \$83.33 among upper quartile rental price units, compared to the control census block groups. These increases represent a rental increase of 6.07 percent among lower quartile rental price units, 7.39 percent among median rental units, and 7.64 percent among upper quartile rental units, respectively.

Columns 4 – 6 present the average treatment effect on the treated of the policy on housing valuations across the lower quartile, median, and upper quartile distribution areas. Unlike for the rental prices, we fail to find statistically significant differences between the treated and control groups at the 10 percent level for lower quartile and high quartile housing valuations, though lower quartile housing valuations is significant. We find positive effects of the policy on housing valuations across the distribution. Lower quartile housing valuations increased by \$11,480, or 6.66 percent compared to the pre-treatment mean. Median housing valuations increased by \$17,384, or 7.01 percent compared to the pre-treatment mean. Finally, upper quartile housing valuations increased by \$12,041, or 3.57 percent compared to the pre-treatment mean.

7 Robustness

7.1 Alternative Matching and Weighting Procedures

The main specification matched treated census block groups in Southern Philadelphia to other census block groups from around Philadelphia based on pre-policy demographics and long-run crime trend variables. The justification for this comparison group was to compare treated census block groups to block groups around Philadelphia that were experiencing similar trends in gun crime incidents, severe part 1 violent crime incidents, and less severe part 2 crime incidents. As a robustness exercise, we follow a similar matching procedure utilizing alternative matching criteria, such as matching pre-policy crime levels (averages), and matching crime levels and trends to test the sensitivity of our results to the matching procedure. In addition, we also test the alternative to the matching procedure by utilizing augmented inverse probability weights which weighs the full sample of all Philadelphia census block groups based on their probability of receiving treatment

based on the same controls in our main specification (i.e. demographics and long-run crime trends).

Table 10 shows the results of this robustness specification exercise across our crime dependent variables of interest. Across all four specifications, the main findings on shootings and crime remain stable and comparable. The intervention continues to be associated with a statistically significant reductions in shooting victims, total crime incidents, gun crime incidents, and total crime incidents, with sizeable effects, albeit varying in size, maintain significance and direction. Shooting victims reductions range from approximately 29% to 41% across specifications, and gun crime incidents range roughly from 11% to 25%. The magnitude of the results decreases with more demanding and restrictive matching procedures. The AIPW weighting procedure also provides comparable results to the mainline specification, providing slightly smaller magnitudes across all categories, but once again is expected given the introduction of census block groups who look non-similar to the treated area.

Table 11 presents the results of these alternative matching and weighting specifications on the downstream results of rental prices and housing markets. For rental price outcomes, the results are reasonably consistent in terms of magnitude and statistical significance. Panel C, which presents the most restrictive matching procedure for matching based on levels and trends, provides smaller and statically noisier estimates of rental prices. Housing value effects are positive in direction but statistically noisier, failing to achieve consistent significance across comparison group constructions and should therefore be interpreted with caution. Taken together Table 10 and 11 suggest that the main findings are not due to a particular comparison group construction. While the crime and downstream effect magnitudes vary predictably with matching or weighting strategy, the direction and significance of the main specification results hold.

8 Discussion

To summarize, we find that Philadelphia’s focused deterrence policy produced a significant reduction in the number of shooting victims and gun crime incidents in treated census block groups compared to matched census block groups from the rest of the city, based on demographics and long-run

crime trends. In addition to gun-related violence, there was a sizable reduction in total crime incidents driven by a reduction in Part 2 crime incidents (less serious, non-violent crimes) rather than reductions in Part 1 crime incidents (violent crimes against persons and property). We further find evidence that this reduction in crime is capitalized into rental prices and housing valuations, as rental prices increase by around 6.0 to 7.6 percent and housing values increase by 3.6 to 7.0 percent post-policy compared to the control census block groups. Overall, these findings suggest that Philadelphia’s focused deterrence policy had a noticeable impact on crime reduction—both in terms of shootings and gun crime incidents and non-violent crime—in treated census block communities, and that rental and housing markets in these communities internalized the reduction in crime into their prices.

The first question one may ask is how these findings, and their magnitude, align with the broader literature on focused deterrence. [Roman et al. \(2019\)](#) finds that the number of shooting victims per 10,000 residents declines by 35 percent in the first 24 months following implementation of Philadelphia’s focused deterrence policy at the census block group level. Our primary specification, which utilizes a different control group, a varied difference-in-differences specification, and a longer time horizon, finds a 41.4 percent reduction in shooting victims. Our alternative specifications yield a range of reductions from 29.1 to 41.4 percent, which aligns closely with the result found in [Roman et al. \(2019\)](#).

While crime incident counts and composition have not previously been explored in the Philadelphia context, the broader focused deterrence literature provides a reasonable benchmark for possible outcomes. Papers that have explored crime incidents as an outcome and found significant effects report reductions in total crime incidents ranging from approximately 15 to 30 percent, and reductions in non-violent offense incidents of around 22 percent ([Saunders et al., 2016](#); [Corsaro et al., 2013](#)). Our results find a reduction in total crime incidents of approximately 14.0 percent in the main specification, ranging from 7.0 to 20.8 percent across alternative specifications, and a reduction in non-violent incidents (Part 2 incidents) of 22.7 percent, ranging from 9.1 to 31.4 percent across alternative specifications. These results again fall reasonably within the range of crime reductions documented in the existing focused deterrence literature.

Our estimates allow for a discussion of the partial benefits of the focused deterrence policy. A complete cost-benefit analysis would require more direct estimates for both program benefits and implementation and operational costs. While precise cost data for Philadelphia’s focused deterrence program is not currently publicly available, the literature on this program and FD programs more generally suggest that the primary costs of the program consisted of a small number of staff salaries, specifically a community manager and social service manager, and coordination costs between different agencies involved in the program, suggestive of relatively low costs of implementation ([Roman et al. \(2019\)](#), [Braga et al. \(2017\)](#)). Given the absence of a precise total cost figure, we focus on our discussion of the potential benefits of the policy. We provide back-of-the-envelope, conservative lower-bound estimates of program benefits through two complementary frameworks: a cost-of-crime approach and a revealed preference framework using rental and housing market outcomes.

We begin with the cost-of-crime approach, which translates our estimated crime reductions into avoided societal costs using per-incident cost estimates from [McCollister et al. \(2010\)](#), who estimate the societal costs of crime via a cost-of-illness and jury compensation approach across several major crime categories. For this approach, we utilize the shooting victims estimations, which measure the reduction in the number of shooting victims as opposed to the reduction of shooting incidents. Our estimates find a reduction of 0.373 shooting victims per treated census block group, against a pre-treatment mean of 0.902, yielding an avoided 0.337 shooting victims per block group per year. Across the 143 treated census block groups in our sample, this implies approximately 48.2 shooting victims avoided annually due to the policy. While [McCollister et al. \(2010\)](#) does not provide a direct cost estimate for gunshot victimization, we use the societal costs of murder and aggravated assault as bounds. Drawing on the shooting victim dataset, in which approximately 19.4 percent of shooting victims were fatal over the course of the sample, and assuming the policy-induced reduction maintained this same fatality rate, the avoided shooting victims decompose into approximately 9.4 murders and 38.8 nonfatal assault victims per year. Applying societal costs of \$8,982,907 per murder and \$107,020 per aggravated assault yields a total avoided societal cost of approximately \$88.6 million per year from shooting reductions alone. While our results also indicate significant

reductions in the number gun crime incidents and Part 2 crime incidents, the heterogeneity of offense types within these categories, the variation in societal costs across individual offense types, and the inability to identify policy-induced responses at the offense level prevent a cost-of-crime estimate for these additional crime reductions.

A complementary approach to estimating the benefits of focused deterrence draws on our rental and housing market estimates through a revealed preference framework. To the extent that rental and housing markets reflect the desirability of a neighborhood, the capitalization of crime reductions into rental and housing prices reflects the marginal willingness to pay of residents and property owners for the bundle of amenity improvements produced by the policy, in this case the reduction of shootings, and other less serious crime offenses. Our estimates find that rental prices increased by approximately 6.0 to 7.6 percent and housing values increased by approximately 3.6 to 7.0 percent in treated census block groups relative to control block groups following the policy. Using the midpoints of these ranges and pre-treatment median values of \$799.50 for rent and \$229,732.20 for housing valuations, we can translate these estimates into aggregate dollar figures across the number of rental units and owner-occupied housing units in our treated group. These quantity of these rental units and owner-occupied housing units are estimated from the ACS 5-year data, and the number of units was estimated in 2012 (one-year prior to the implementation of the policy). For Southern Philadelphia, the 2012 ACS 5-year estimates suggest that there are around 30,255 renter-occupied units and 38,940 owner-occupied units. The estimated rental price increase implies approximately \$18.9 million in additional annual rental market value across treated communities, while the estimated housing value increase implies approximately \$474.1 million in aggregate housing wealth gains. As these figures reflect only the portion of policy benefits captured by local property markets, they should be interpreted as a conservative lower bound on the true welfare gains from the policy.

Taken together, these estimates suggest that Philadelphia’s focused deterrence policy was highly cost-effective. Even using only the cost-of-crime framework and only the reduction in shooting victims, the policy generated approximately \$ 88.6 million in avoided societal costs per year, a figure that far exceeds any plausible estimate of the program’s operating costs given its reliance on

a small number of staff positions and interagency coordination ([Roman et al., 2019](#); [Braga et al., 2017](#)). The revealed preference estimates point in the same direction, with the capitalization of crime reductions into rental and housing markets implying substantial gains in residents’ willingness to pay for safer neighborhoods. Importantly, the favorable balance of benefits and costs does not depend on crime reductions as large as those we observe. Given the combined fatality rate in our sample, each avoided shooting victim corresponds to approximately \$ 1.8 million in avoided societal costs, so a program with annual operating costs on the order of \$1 million would need to prevent only one shooting victim per year to generate positive net benefits. This implies that focused deterrence may yield positive net benefits even in settings where the absolute reduction in violence is modest, such as smaller cities or jurisdictions with lower baseline levels of gun violence. The low and largely fixed costs of implementation suggest that the policy is best suited to contexts where violence is concentrated among identifiable groups and where the agencies needed for coordination already exist, conditions that are common across many American cities regardless of size.

9 Conclusion

This study examines the effect of Philadelphia’s focused deterrence strategy on crime and on how those crime reductions are capitalized into local housing markets. Using a difference-in-differences design with propensity score matching at the census block group level, we find that FD reduced shooting victims by 41.3 percent and gun crime incidents by 24.2 percent in treated communities. Total reported crime fell by roughly 13.8 percent, a decline driven largely by reductions in less serious non-violent offenses, suggesting that the policy has effects that extended beyond just violent. These increases in public safety were capitalized into local housing markets, with rental prices rising between 6.0 to 7.5 percent and housing valuations rising between 3.6 to 7.0 percent in treated areas in the years following implementation.

These findings carry several implications for policymakers considering the effectiveness of focused deterrence strategies. First, these results add to the existing evidence that FD can produce large and sustained reductions in serious gun violence. Our post-period extends well beyond the

short evaluation windows used in prior studies of the Philadelphia program, and the declines in shootings and gun crime persist throughout. This suggests that FD’s effects are not merely a short-lived response to increased enforcement attention. Second, the reductions in less serious non-violent crime indicate that the policy’s impact reaches beyond just its target on gang-related gun violence. Whether these reductions are attributed to complimentary declines in other forms of crime by targeted individuals, or by broader changes to community norms, reduced retaliatory cycles, or shifts in police and community attention is a subject of future research. These results imply that evaluations focused solely on targeted violent outcomes may understate the program’s total benefits. Third, both the social cost of crime and revealed-preference framework provide a methodology to gauge how much residents value improved public safety by the program. Together, these market responses suggest that the returns to focused deterrence are substantial relative to its costs, strengthening the case for this type of program as a cost-effective tool for cities facing concentrated gun violence.

While we measure downstream effects primarily through rental and housing markets, reductions in violent crime are likely to shape community outcomes through other channels as well. Future research should examine effects on commercial property values, business formation, and broader economic activity. In addition, this environment would be useful to other outcomes such as educational outcomes such as student achievement and school attendance, where exposure to violence has well-documented consequences. Further work on ultimately bears the incidence of rising housing and rental costs, whether through tenant turnover or evictions, and on the durability of FD’s effects over longer horizons and across other implementations, would help clarify the full cost-benefits of the FD policy.

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A Tables and Figures

A.1 Summary Statistics

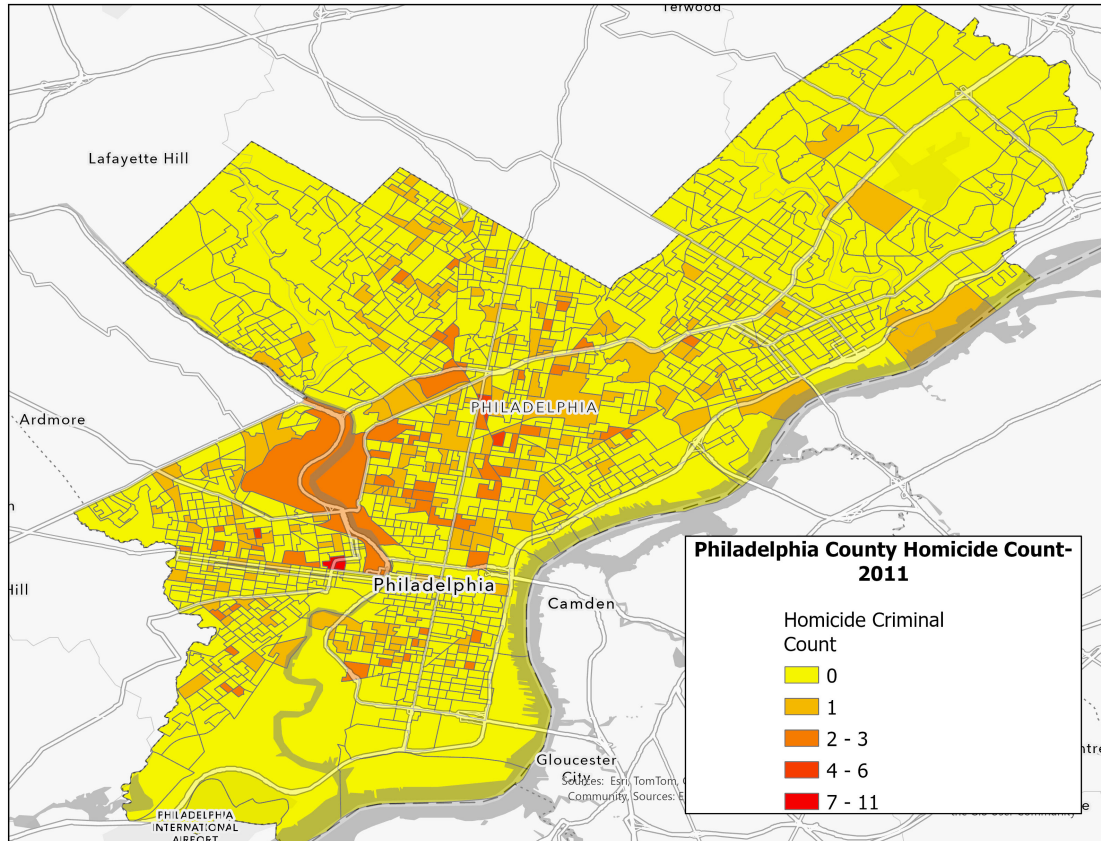


Figure 1: Philadelphia Homicide Crime Incidents, 2011

Notes: Homicide incident data are drawn from the Philadelphia Police Department's incident-level crime dataset. An incident is a reported offense in which one offender, or group of offenders acting in concert, commits an offense at the same time and place. The map reports the number of criminal homicide incidents, defined as murder and nonnegligent manslaughter; negligent deaths, attempted killings, assaults with intent to kill, suicides, and accidental deaths are excluded. Incidents are geocoded and aggregated to 2010 census block groups. The map displays counts for calendar year 2011.

Table 1: Pre-Treatment (2010-2012) Crime and Community Characteristics Means

	(0) South Philadelphia	(1) Rest of Philadelphia	Difference (0) – (1)
<i>Panel A: Shootings and Crime Incidents</i>			
Shooting Victims	0.929 (0.082)	1.090 (0.030)	–0.161* (0.087)
Total Crime Incidents	134.505 (4.399)	140.233 (2.271)	–5.729 (4.951)
Part I Crime Incidents	50.299 (1.766)	54.855 (1.074)	–4.556** (2.067)
Part II Crime Incidents	84.205 (3.066)	85.378 (1.434)	–1.173 (3.384)
Gun Crime	4.416 (0.200)	5.933 (0.099)	–1.517*** (0.223)
<i>Panel B: Community Characteristics</i>			
Population	1123.822 (23.930)	1139.123 (8.841)	–15.301 (25.511)
% Black	0.293 (0.016)	0.494 (0.006)	–0.201*** (0.017)
% Hispanic	0.060 (0.004)	0.121 (0.003)	–0.061*** (0.005)
% Bachelor’s or above	0.284 (0.011)	0.203 (0.003)	0.081*** (0.012)
Labor Force Participation	0.631 (0.006)	0.588 (0.002)	0.043*** (0.007)
Unemployment Rate	0.067 (0.003)	0.086 (0.001)	–0.019*** (0.003)
Share of Households on SNAP	0.110 (0.006)	0.143 (0.003)	–0.033*** (0.007)
Share of Households below Poverty Line	0.182 (0.008)	0.215 (0.003)	–0.033*** (0.009)
Median Household Income	\$42,275.61 (1026.760)	\$38,486.80 (350.093)	+\$3,788.81*** (1085.025)
Observations	438	3,546	3,984

Notes: Crime incident and shooting victim data are drawn from the Philadelphia Police Department’s crime incident and shooting victim dataset. Part I crime incidents are defined as serious and violent crimes against persons and property and include incidents such as criminal homicide, aggravated assault, robbery, burglary, and motor vehicle theft. Part II crime incidents are less serious, typically non-violent offenses and include incidents such as fraud, forgery, embezzlement, buying and possessing stolen property, and gambling. Gun crime incidents are incidents involving a firearm and include homicide, aggravated assault with a firearm, robbery with a firearm, and weapon violations. Community characteristics originate from American Community Survey 5-year estimates. All measures reflect average annual values for 2010–2012 at the census block group level, prior to the start of the Focused Deterrence intervention. South Philadelphia represents the census block groups defined as those receiving the Philadelphia Focused Deterrence treatment, as stated in [Roman et al. \(2017\)](#); this is primarily defined as census block groups falling under the jurisdiction of Police Districts 1, 3, and 17. All other census block groups in Philadelphia County represent the Rest of Philadelphia category. Column (0) – (1) reports the difference in means, with standard errors in parentheses.

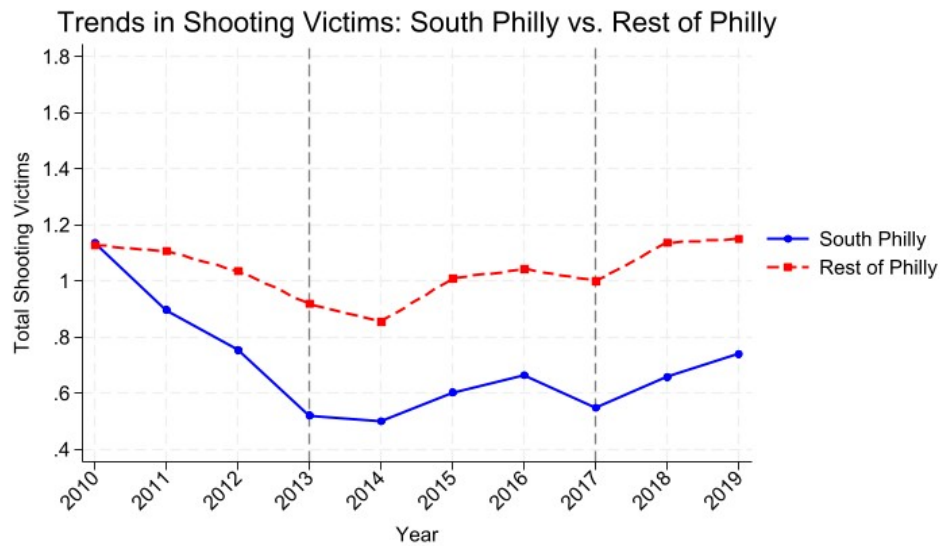
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Pre-Treatment Means (2010-2012): Rental Prices and Housing Valuations

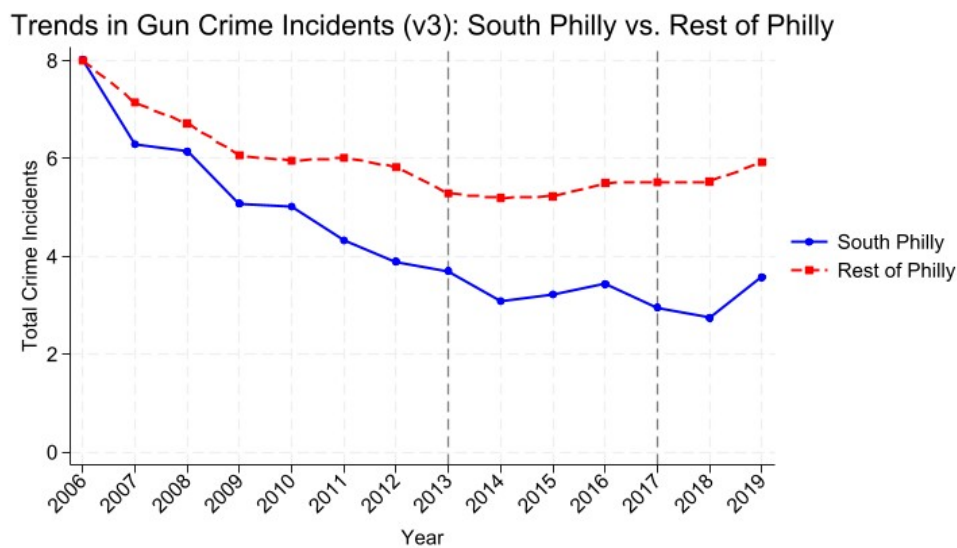
	(1) South Philadelphia	(0) Rest of Philadelphia	Difference (1) – (0)
Low Qrt. Rent	\$619.83 (11.165)	\$609.25 (4.334)	\$10.58 (11.977)
Median Rent	\$799.50 (12.768)	\$774.53 (4.761)	\$24.97* (13.626)
High Qrt. Rent	\$1,026.50 (17.824)	\$946.79 (5.965)	\$79.71*** (18.796)
Low Qrt. Housing Valuation	\$157,426.80 (5412.605)	\$120,664.50 (1687.135)	\$36,762.30*** (5669.455)
Median Housing Valuation	\$229,732.20 (7328.493)	\$165,754.00 (2284.047)	\$63,978.20*** (7676.176)
High Qrt. Housing Valuation	\$315,245.10 (10251.980)	\$221,073.50 (3098.045)	\$94,171.60*** (10709.854)
Observations	410	3,253	3,577

Rent and housing valuation data originates from American Community Survey 5-year census block group estimates. Rental prices represent the household's monthly contract rent payment in 2019 dollars. Housing valuation data represents self-reported valuation of how much a house and lot would sell for if it was for sale in 2019 dollars. All measures reflect average annual values for 2010–2012 at the census block group level, prior to the start of the Focused Deterrence intervention. South Philadelphia represents the census block groups defined as those receiving the Philadelphia Focused Deterrence treatment, as stated in [Roman et al. \(2017\)](#); this is primarily defined as census block groups falling under the jurisdiction of Police Districts 1, 3, and 17. All other census block groups in Philadelphia County represent the Rest of Philadelphia category. Column (0) – (1) reports the difference in means, with standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$



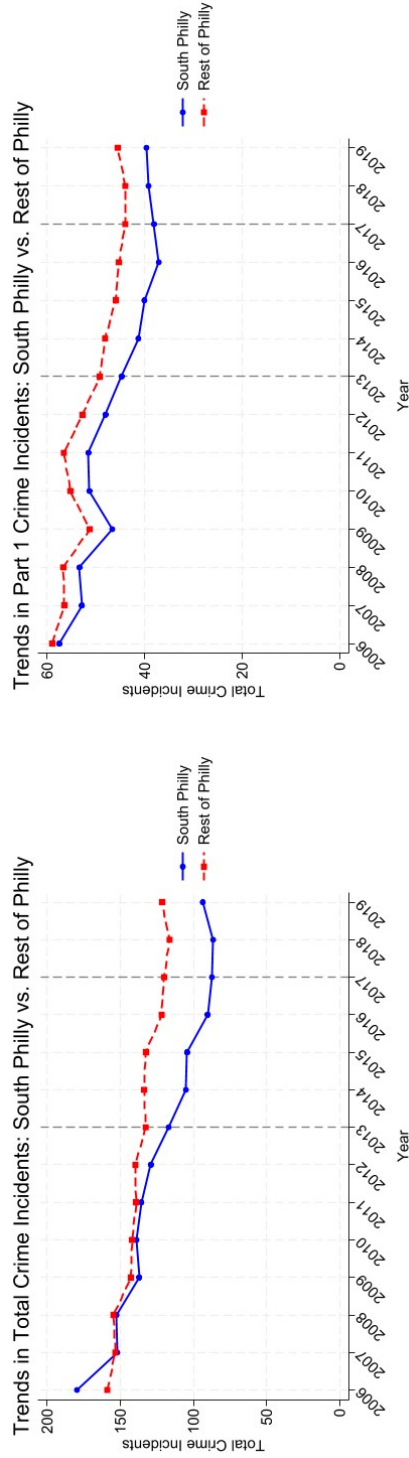
(a) Average Shooting Victims (2010–2019)



(b) Average Gun Crime Incidents

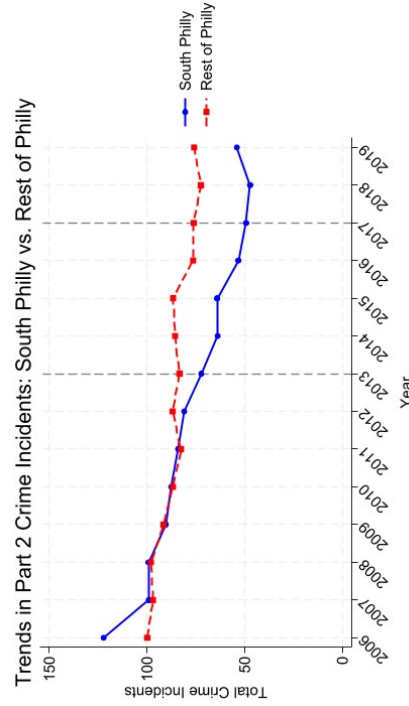
Figure 2: Pre-Matched Average Shooting Victims and Gun Crime Incidents, 2006–2019

Notes: Shooting victim and crime incident data are drawn from the Philadelphia Police Department's crime incident and shooting victim dataset. Gun crime incidents are incidents involving a firearm and include homicide, aggravated assault with a firearm, robbery with a firearm, and weapon violations. South Philadelphia represents the census block groups defined as those receiving the Philadelphia focused deterrence treatment, as stated in ?; this is primarily defined as census block groups falling under the jurisdiction of Police Districts 1, 3, and 17. All other census block groups in Philadelphia County represent the Rest of Philadelphia category. Series report raw, unmatched annual means by group prior to any covariate matching procedure. Dashed vertical lines indicate start of FD policy in 2013, and end of policy in 2017.



(a) Average Total Crime Incidents

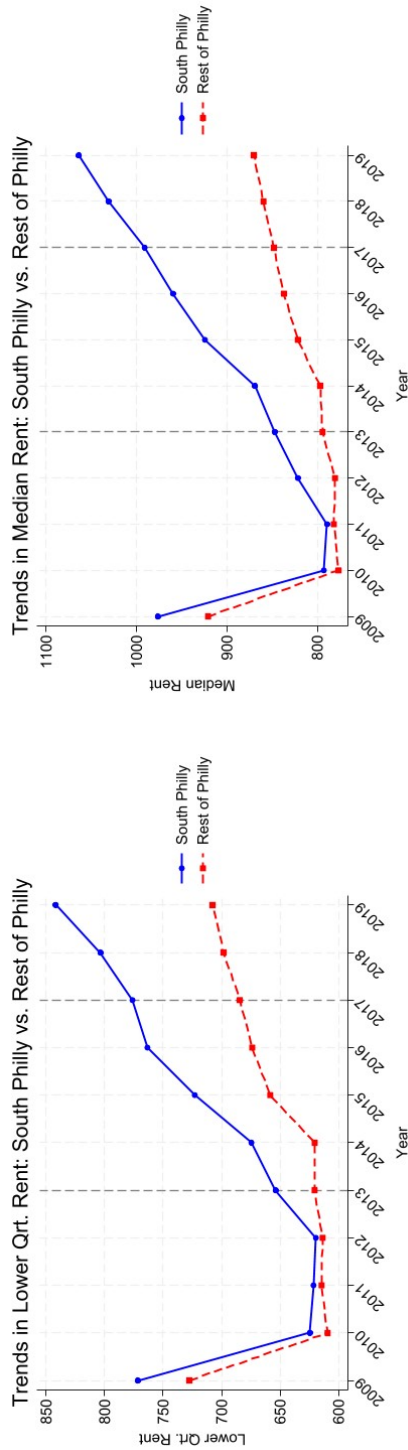
(b) Average Part 1 Crime Incidents



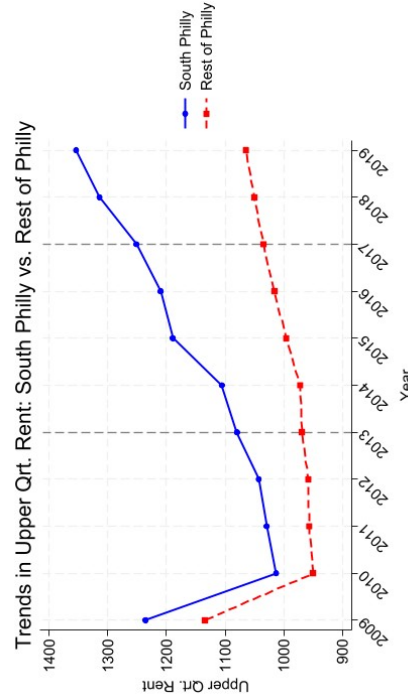
(c) Average Part 2 Crime Incidents

Figure 3: Pre-Matched Average Total, Part I, and Part II Crime Incidents, 2006–2019

Notes: Crime incident data are drawn from the Philadelphia Police Department's crime incident dataset. Part I crime incidents are defined as serious and violent crimes against persons and property and include incidents such as criminal homicide, aggravated assault, robbery, burglary, and motor vehicle theft. Part II crime incidents are less serious, typically non-violent offenses and include incidents such as fraud, forgery, embezzlement, buying and possessing stolen property, and gambling. South Philadelphia represents the census block groups defined as those receiving the Philadelphia Focused Deterrence treatment, as stated in ?; this is primarily defined as census block groups falling under the jurisdiction of Police Districts 1, 3, and 17. All other census block groups in Philadelphia County represent the Rest of Philadelphia category. Series report raw, unmatched annual means by group prior to any covariate matching procedure. Dashed vertical lines indicate start of FD policy in 2013, and end of policy in 2017.



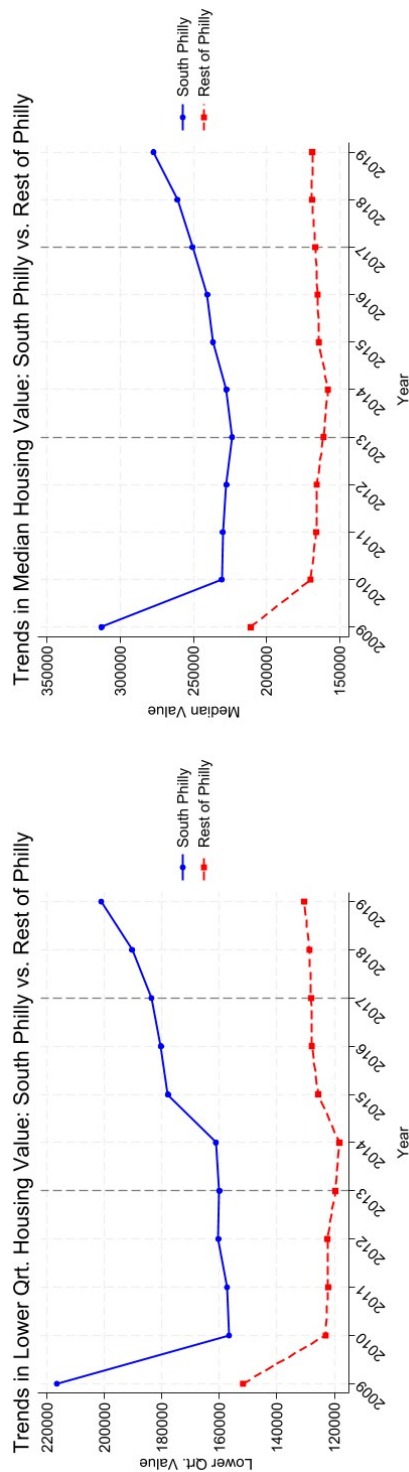
(a) Average Lower Qrt. Rental Prices



(b) Average Median Rental Prices

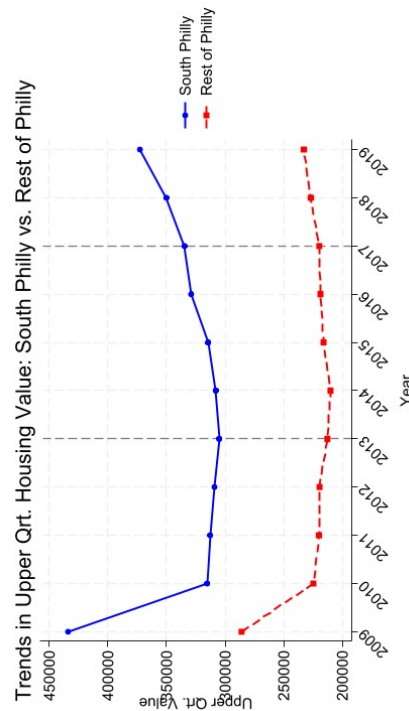
(c) Average High Qrt. Rental Prices

Notes: Rental price data originate from American Community Survey 5-year census block group estimates. Rent represents the household's monthly contract rent payment, reported in 2019 dollars. Lower Qrt., Median, and Upper Qrt. Rent report the 25th percentile, 50th percentile, and 75th percentile of contract rent, respectively, within each group-year. South Philadelphia represents the census block groups defined as those receiving the Philadelphia focused deterrence treatment, as stated in [Roman et al. \(2017\)](#); this is primarily defined as census block groups falling under the jurisdiction of Police Districts 1, 3, and 17. All other census block groups in Philadelphia County represent the Rest of Philadelphia category. Series report raw, unmatched annual means by group prior to any covariate matching procedure. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017.



(a) Average Low Qrt. Housing Valuations

(b) Average Median Housing Valuations



(c) Average High Qrt. Housing Valuations

Figure 5: Pre-Matched Average Housing Valuations

Notes: Housing valuation data originate from American Community Survey 5-year census block group estimates. Housing valuation represents the household's self-reported estimate of how much their house and lot would sell for if it were for sale, reported in 2019 dollars. Lower Qtr., Median, and Upper Qtr. Valuation report the 25th percentile, 50th percentile, and 75th percentile of housing valuation, respectively, within each group-year. South Philadelphia represents the census block groups defined as those receiving the Philadelphia focused deterrence treatment, as stated in [Roman et al. \(2017\)](#); this is primarily defined as census block groups falling under the jurisdiction of Police Districts 1, 3, and 17. All other census block groups in Philadelphia County represent the Rest of Philadelphia category. Series report raw, unmatched annual means by group prior to any covariate matching procedure. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017.

A.2 PSM - Matching - Trends in Crime

Table 3: PSM Covariate Balance

Characteristic	Unmatched				Matched			
	Treated	Comp.	% Bias	<i>t</i> -stat.	Treated	Comp.	% Bias	<i>t</i> -stat.
<u>Crime Trends (2006–2012):</u>								
Part 1 Crime Count Trend	−9.466	−6.633	−14.4	−1.49	−8.867	−7.287	−8.0	−0.59
Part 2 Crime Count Trend	−41.212	−10.461	−41.4	−5.77	−33.462	−23.294	−13.7	−1.91
Gun Crime Count Trend	−4.123	−2.034	−37.4	−4.71	−3.881	−2.895	−17.7	−1.56
<u>Community Characteristics (Avg. 2009–2012):</u>								
Youth Male Density	0.065	0.077	−24.0	−2.34	0.065	0.070	−8.3	−0.73
Total Population	1133.5	1160.9	−5.3	−0.58	1136.6	1167.8	−6.1	−0.50
Percent Black	0.296	0.471	−49.2	−5.34	0.299	0.221	21.9	2.07
Percent Hispanic	0.060	0.124	−43.0	−3.88	0.060	0.052	5.4	0.88
Econ. Disadvantage Index	−0.228	−0.005	−29.9	−3.22	−0.219	−0.242	3.1	0.27
Number of Census Block Groups	146	1,182			143	119		

Notes: Balance statistics computed via `pstest` following nearest-neighbor propensity score matching with replacement. Matching procedure found the 1 nearest neighbor with a caliper of 0.03. % Bias is the standardized percentage bias in means between treated and comparison units. Rubin’s $B = 40.1$ for the matched sample. Three treated block groups were dropped due to lack of common support.

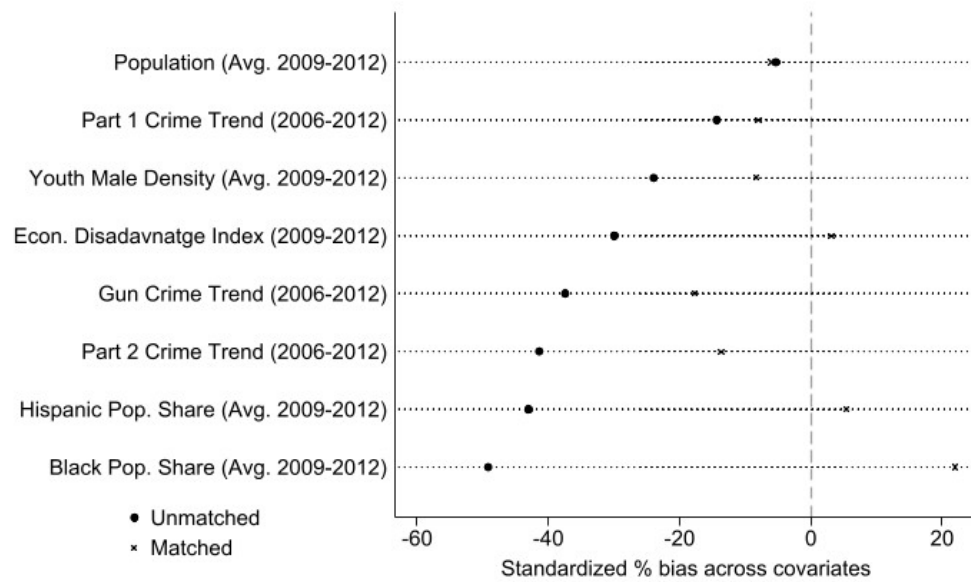
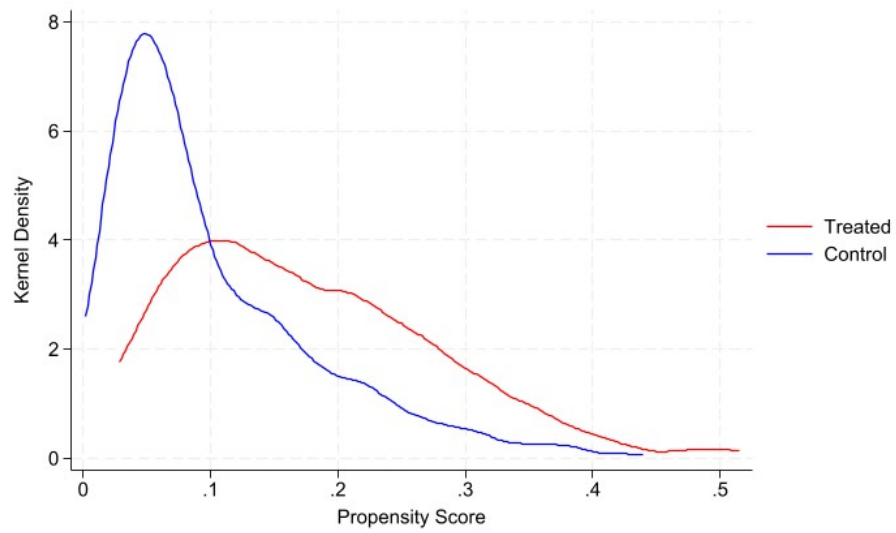
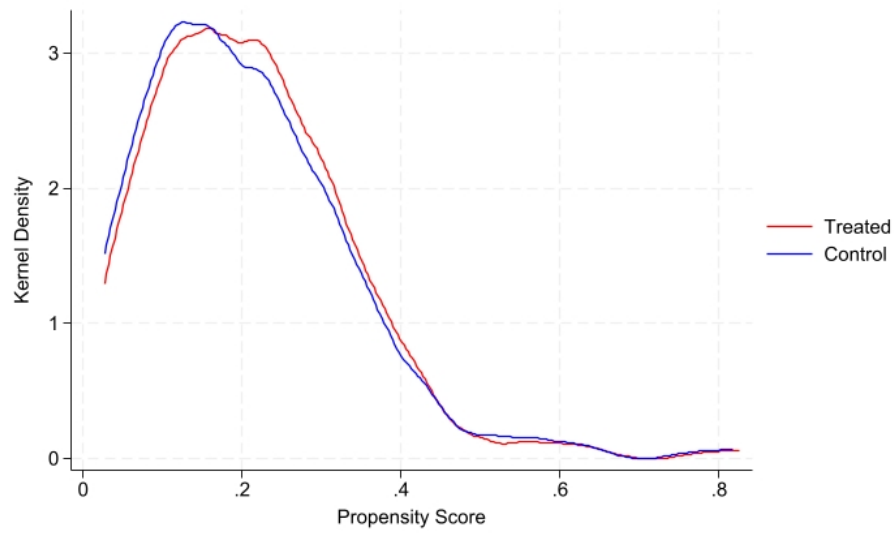


Figure 6: Covariate Balance Loveplot

Notes: Standardized % bias across covariates computed via `pstest` before (Unmatched) and after (Matched) nearest-neighbor propensity score matching with replacement. The dashed vertical line at zero indicates no standardized bias between treated and comparison units. See Table 3 for underlying covariate balance statistics and matching procedure details.



Pre-Match



Post-Match

Figure 7: Kernel Density of Propensity Scores

Notes: Kernel density estimates of the estimated propensity score for treated and comparison census block groups, before (Pre-Match) and after (Post-Match) nearest-neighbor propensity score matching with replacement. Propensity scores are estimated via a logistic regression on the covariates listed in Table 3.

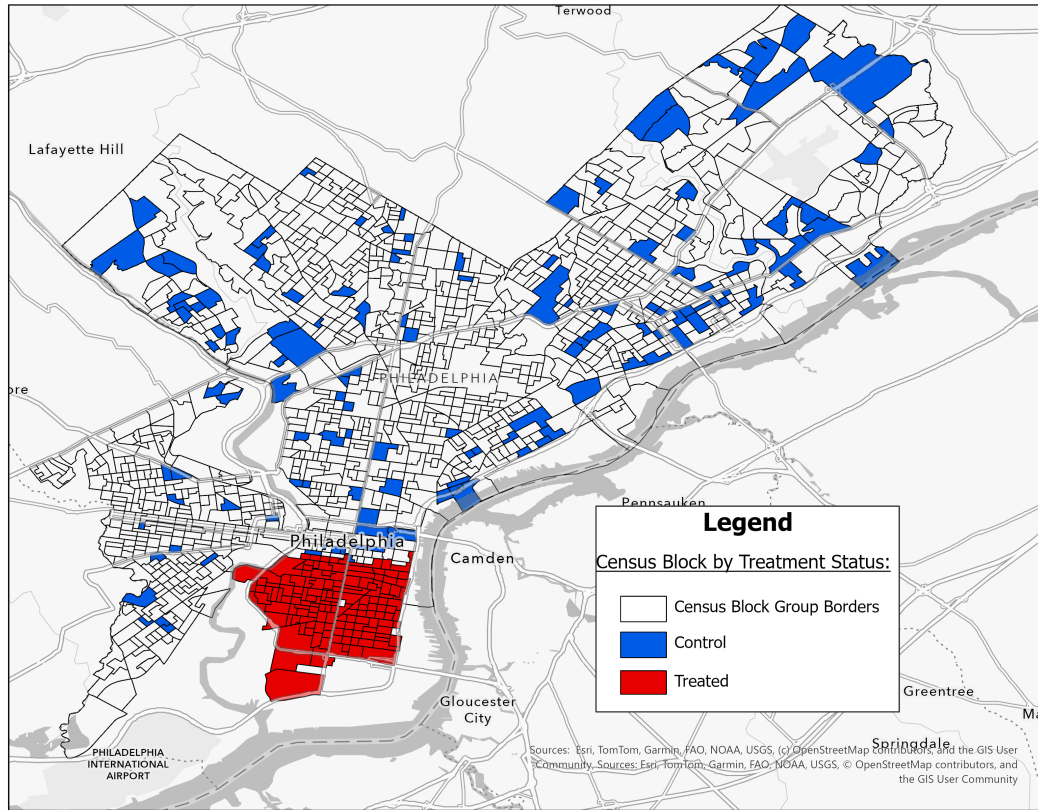


Figure 8: PSM Matched Treatment Map

Notes: Treated census block groups (South Philadelphia) are defined as those receiving the Philadelphia focused deterrence treatment, as stated in [Roman et al. \(2017\)](#); this is primarily defined as census block groups falling under the jurisdiction of Police Districts 1, 3, and 17. Control census block groups represent matched comparison units selected via nearest-neighbor propensity score matching with replacement. See [Table 3](#) for the covariates used in the propensity score estimation.

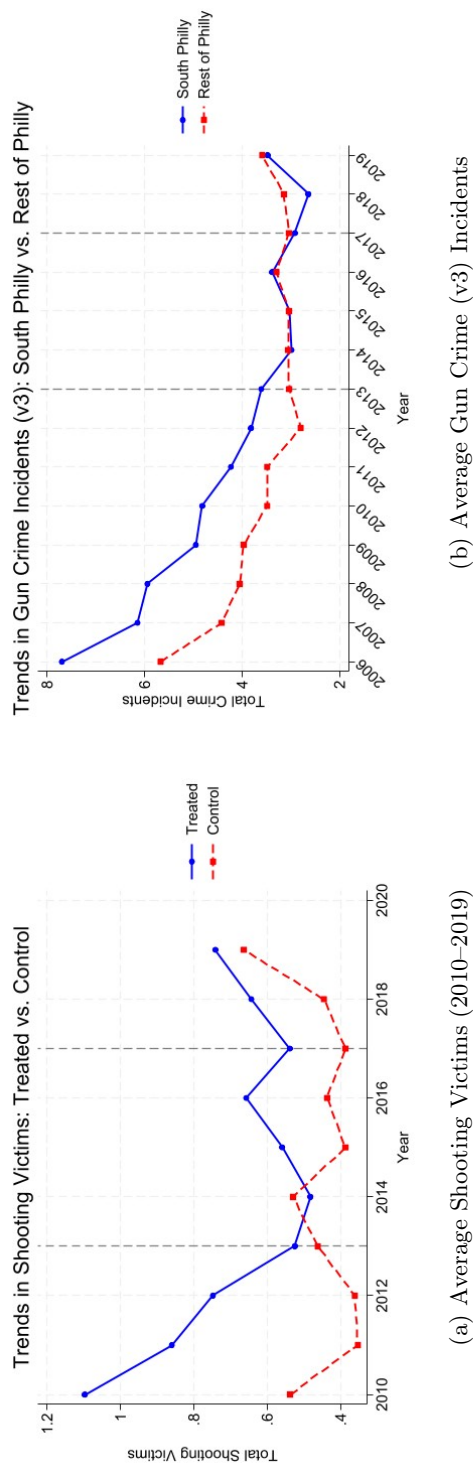
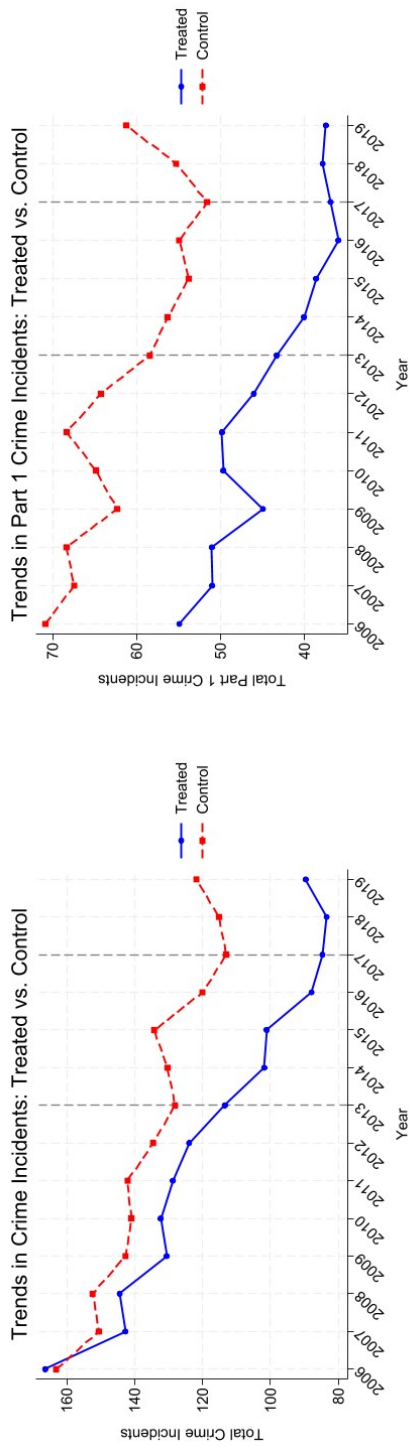
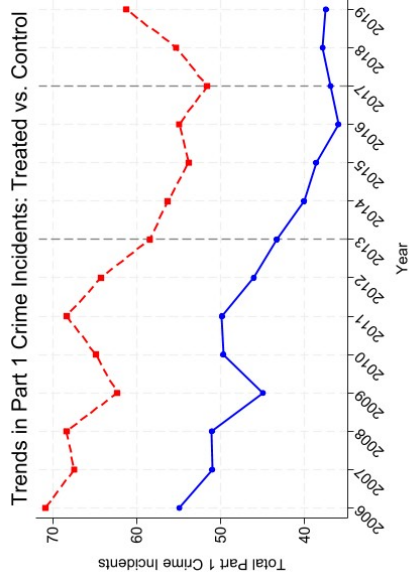


Figure 9: Matched Average Shooting Victims and Gun Crime Incidents, 2006–2019

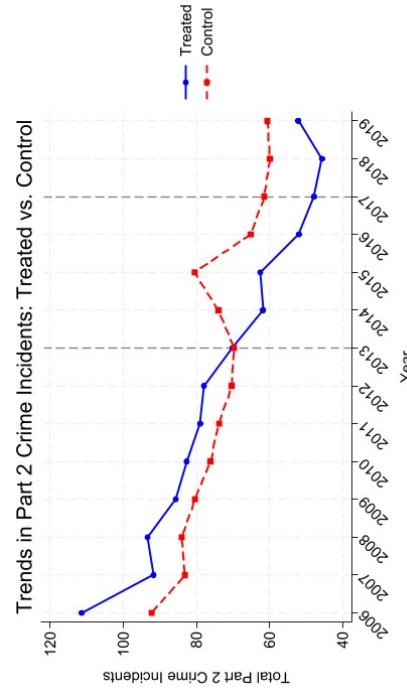
Notes: Shooting victim and crime incident data are drawn from the Philadelphia Police Department’s crime incident and shooting victim dataset. Series report annual means by group following nearest-neighbor propensity score matching with replacement; see Table 3 for the covariates used in the propensity score estimation and matching procedure. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017.



(a) Average Total Crime Incidents



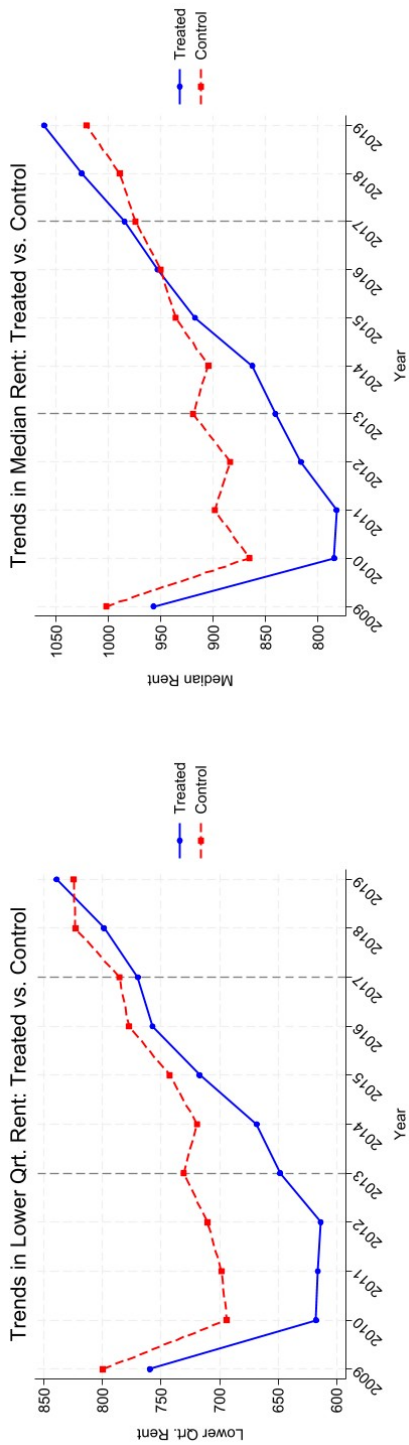
(b) Average Part 1 Crime Incidents



(c) Average Part 2 Crime Incidents

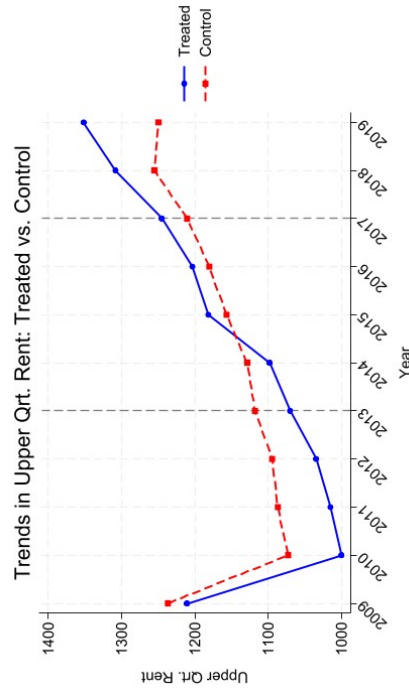
Figure 10: Matched Average Total, Part I, and Part II Crime Incidents, 2006–2019

Notes: Crime incident data are drawn from the Philadelphia Police Department's crime incident dataset. Part I crime incidents are defined as serious and violent crimes against persons and property and include incidents such as criminal homicide, aggravated assault, robbery, burglary, and motor vehicle theft. Part II crime incidents are less serious, typically non-violent offenses and include incidents such as fraud, forgery, embezzlement, buying and possessing stolen property, and gambling. Series report annual means by group following nearest-neighbor propensity score matching with replacement; see Table 3 for the covariates used in the propensity score estimation and matching procedure. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017.



(a) Matched - Average Lower Qrt. Rental Prices

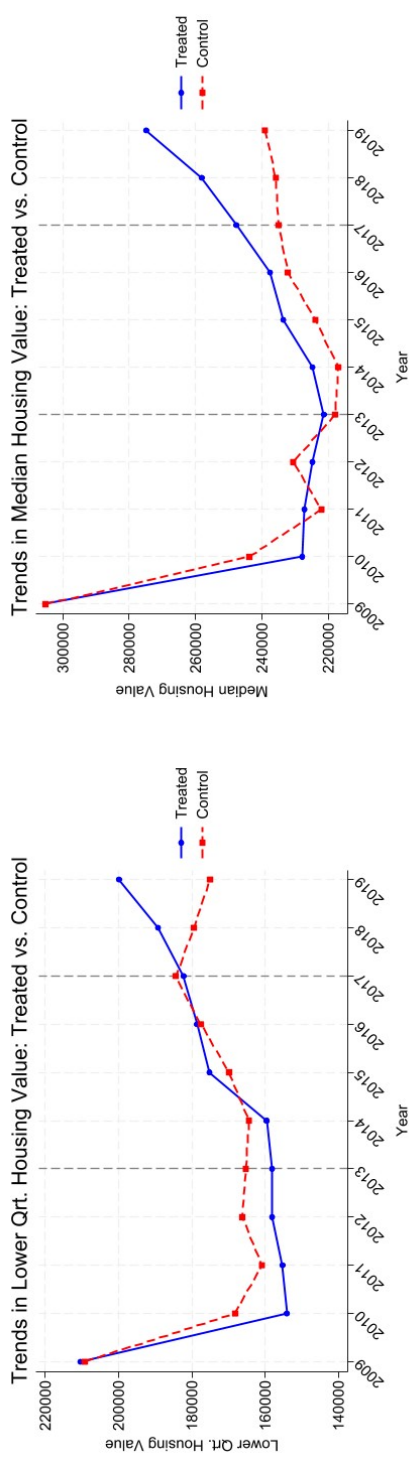
(b) Matched - Average Median Rental Prices



(c) Matched - Average High Qrt. Rental Prices

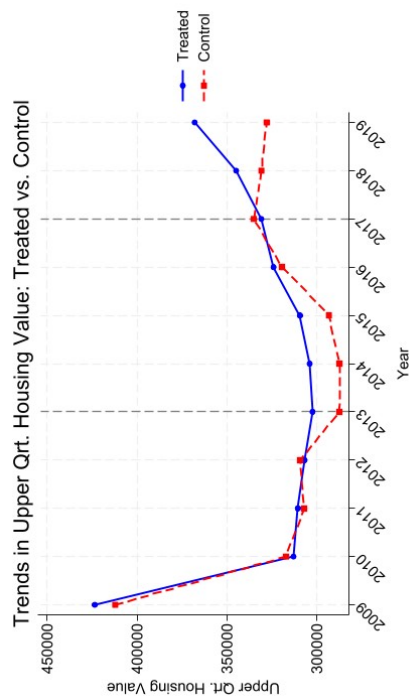
Figure 11: Matched Average Rental Prices

Notes: Rental price data originate from American Community Survey 5-year census block group estimates. Rent represents the household's monthly contract rent payment, reported in 2019 dollars. Lower Qtr., Median, and Upper Qtr. Rent report the 25th percentile, 50th percentile, and 75th percentile of contract rent, respectively, within each group-year. Series report annual means by group following nearest-neighbor propensity score matching with replacement; see Table 3 for the covariates used in the propensity score estimation and matching procedure. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017.



(a) Matched - Average Low Qrt. Housing Valuations

(b) Matched - Average Median Housing Valuations



(c) Matched - Average High Qrt. Housing Valuation

Figure 12: Matched Average Housing Valuations

Notes: Housing valuation data originate from American Community Survey 5-year census block group estimates. Housing valuation represents the household's self-reported estimate of how much their house and lot would sell for if it were for sale, reported in 2019 dollars. Lower Qtr., Median, and Upper Qtr. Valuation report the 25th percentile, 50th percentile, and 75th percentile of housing valuation, respectively, within each group-year. Series report annual means by group following nearest-neighbor propensity score matching with replacement; see Table 3 for the covariates used in the propensity score estimation and matching procedure. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017.

A.3 Results - Event Studies and DiD

Table 4: Crime Raw Means by Treatment Status and Period

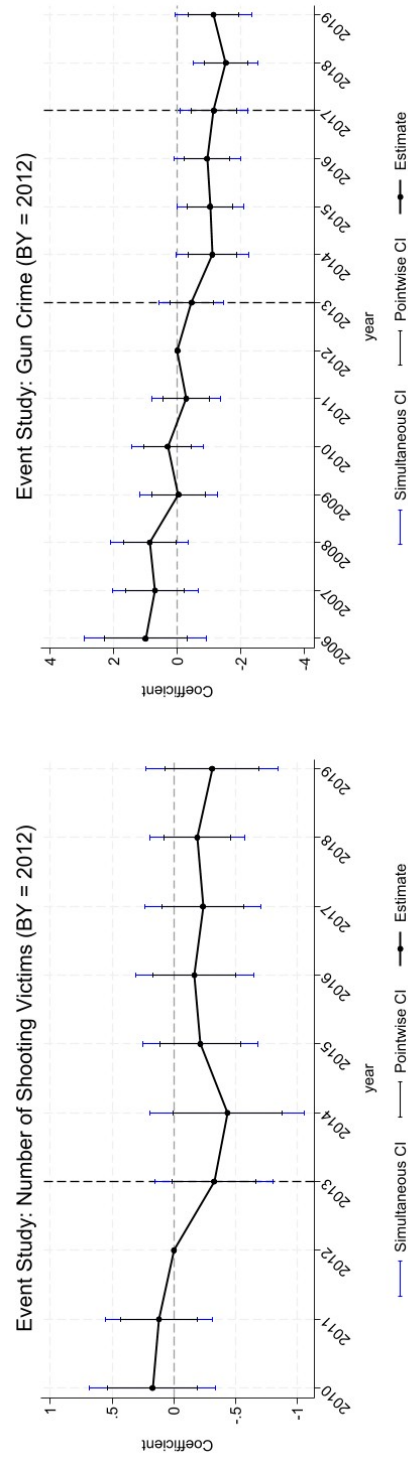
	Control		Treated	
	Pre	Post	Pre	Post
Shooting Victims	0.179	0.473	0.387	0.592
Total Crime	146.61	123.23	138.43	94.58
Part 1 Crime	66.61	55.92	49.62	38.58
Part 2 Crime	79.99	67.31	88.81	56.00
Gun Crime	3.983	3.172	5.373	3.150

Notes: Crime incident and shooting victim data are drawn from the Philadelphia Police Department’s crime incident and shooting victim dataset. Raw means computed on the matched sample. Pre-treatment period is 2010–2012 for shooting victims and 2006–2012 for crime incidents, reflecting differences in data availability by outcome; post-treatment period is 2013–2019 for all outcomes. FD treatment began in 2013.

Table 5: Raw Means by Treatment Status and Period: Rental Prices and Housing Valuations

	Control		Treated	
	Pre	Post	Pre	Post
Low Qrt. Rent	726.01	771.45	651.93	740.14
Median Rent	913.24	957.06	835.71	948.56
High Qrt. Rent	1,123.30	1,186.40	1,065.60	1,208.16
Low Qrt. Housing Val.	176,051	173,553	169,537	177,459
Median Housing Val.	250,280	228,649	246,398	242,500
High Qrt. Housing Val.	336,164	311,539	338,636	326,019

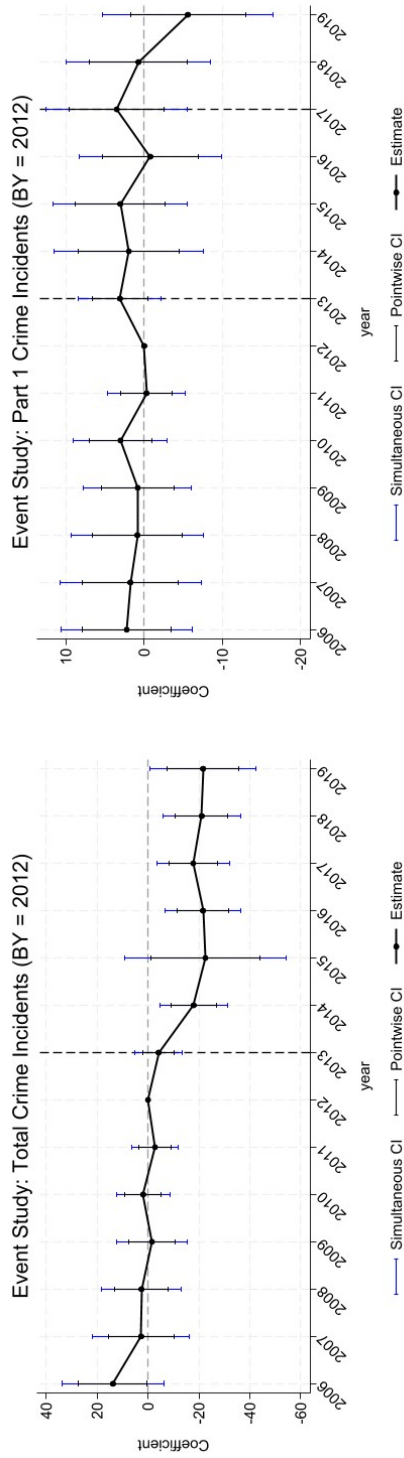
Notes: Rental price and housing valuation data originate from American Community Survey 5-year estimates. Raw means computed on the matched sample. Pre-treatment period is 2009–2012; post-treatment period is 2013–2019. Treatment began in 2013. Rents and housing valuations in 2019 dollars.



(a) Shooting Victims

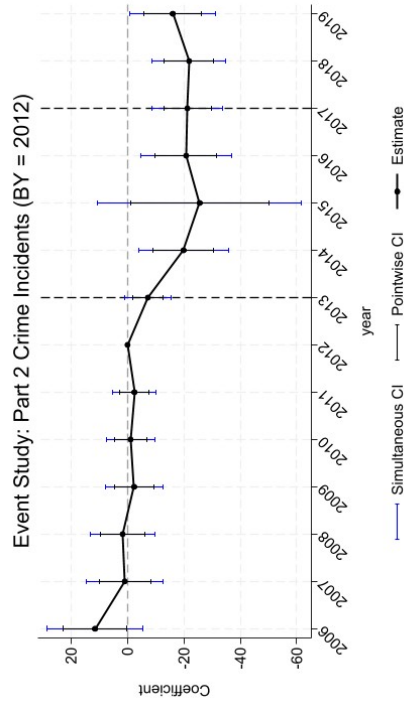
(b) Gun Crime Incidents

Notes: Shooting victim and crime incident data are drawn from the Philadelphia Police Department's crime incident and shooting victim dataset. Gun crime incidents are incidents involving a firearm and include homicide, aggravated assault with a firearm, robbery with a firearm, and weapon violations. Estimates are from an event study specification estimated on the matched sample, including census block group and year fixed effects. Standard errors are clustered at the census block group level. Coefficients are estimated relative to a base year of 2012. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017. 95% pointwise and simultaneous confidence intervals are displayed for each coefficient. See Table 3 for the covariates used in the propensity score estimation and matching procedure.



(a) Total Crime Incidents

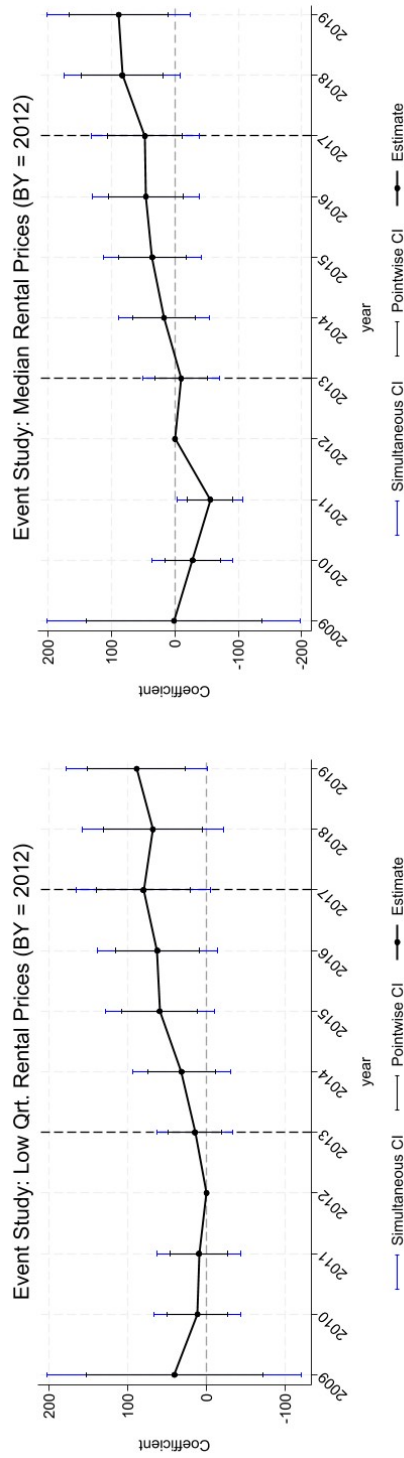
(b) Part 1 Crime Incidents



(c) Part 2 Crime Incidents

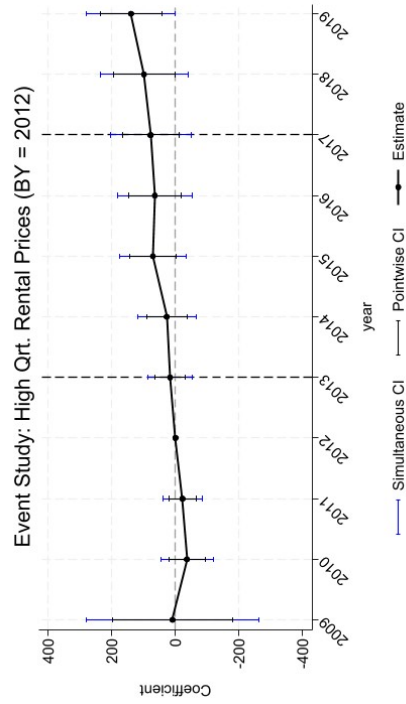
Figure 14: Event Studies – Total, Part I, and Part II Crime Incidents

Notes: Crime incident data are drawn from the Philadelphia Police Department's crime incident dataset. Part I crime incidents are defined as serious and violent crimes against persons and property and include incidents such as criminal homicide, aggravated assault, robbery, burglary, and motor vehicle theft. Part II crime incidents are less serious, typically non-violent offenses and include incidents such as fraud, forgery, embezzlement, buying and possessing stolen property, and gambling. Estimates are from an event study specification estimated on the matched sample, including census block group and year fixed effects. Standard errors are clustered at the census block group level. Coefficients are estimated relative to a base year of 2012. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017. 95% pointwise and simultaneous confidence intervals are displayed for each coefficient. See Table 3 for the covariates used in the propensity score estimation and matching procedure.



(a) Low Qrt. Rent

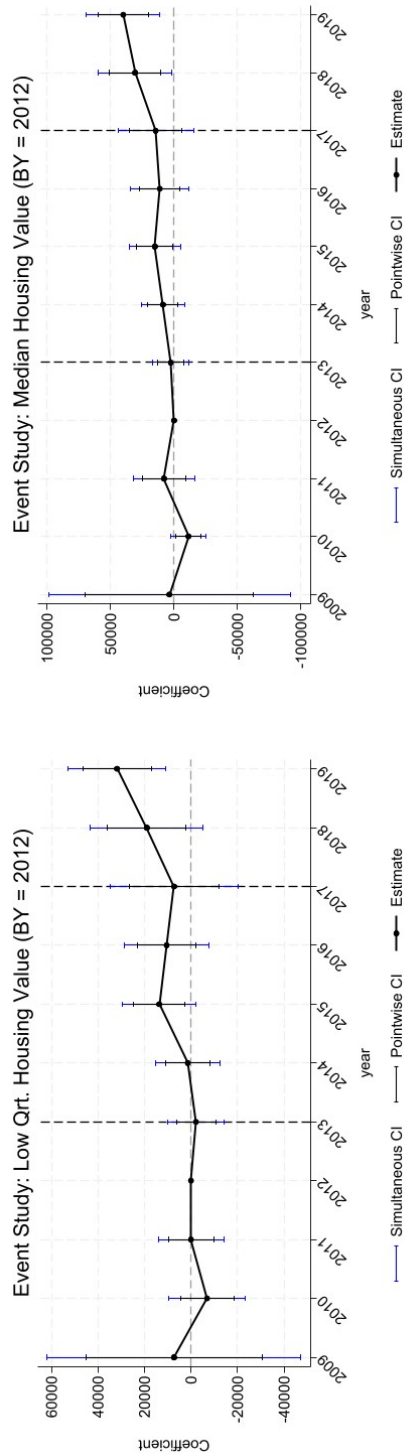
(b) Median Rent



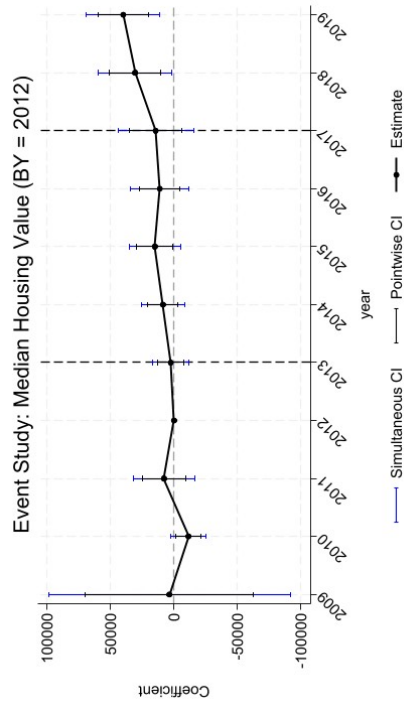
(c) High Qrt. Rent

Figure 15: Event Studies – Rental Prices

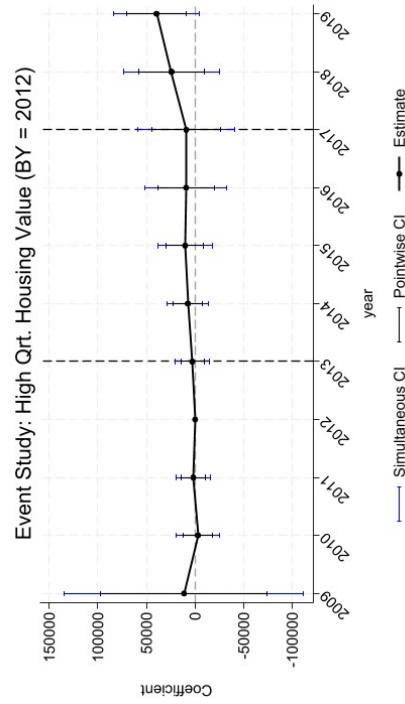
Notes: Rental price data originate from American Community Survey 5-year estimates. Rent represents the household's monthly contract rent payment, reported in 2019 dollars. Low Qrt., Median, and High Qrt. Rent report the 25th percentile, 50th percentile, and 75th percentile of contract rent, respectively. Estimates are from an event study specification estimated on the matched sample, including census block group and year fixed effects. Standard errors are clustered at the census block group level. Coefficients are estimated relative to a base year of 2012. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017. 95% pointwise and simultaneous confidence intervals are displayed for each coefficient. See Table 3 for the covariates used in the propensity score estimation and matching procedure.



(a) Low Qrt. Housing Value



(b) Median Housing Value



(c) High Qrt. Housing Value

Figure 16: Event Studies – Housing Valuations

Notes: Housing valuation data originate from American Community Survey 5-year estimates. Housing valuation represents the household's self-reported estimate of how much their house and lot would sell for if it were for sale, reported in 2019 dollars. Low Qrt., Median, and High Qrt. Valuation report the 25th percentile, 50th percentile, and 75th percentile of housing valuation, respectively. Estimates are from an event study specification estimated on the matched sample, including census block group and year fixed effects. Standard errors are clustered at the census block group level. Coefficients are estimated relative to a base year of 2012. Dashed vertical lines indicate the start of the focused deterrence policy in 2013 and the end of the policy in 2017. 95% pointwise and simultaneous confidence intervals are displayed for each coefficient. See Table 3 for the covariates used in the propensity score estimation and matching procedure.

Table 6: Difference-in-Differences Results: Crime Incidents

	<i>Outcome Variable</i>				
	Shooting Victims	Total Crime	Part I Crime	Part II Crime	Gun Crime
Treat $\times$ Post	-0.373*** (0.091)	-19.165*** (4.489)	0.958 (2.679)	-20.123*** (4.762)	-1.298*** (0.247)
Census Block Group FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES
Pre-Treatment Mean	0.902	138.432	49.619	88.812	5.373
Sample	2010–2017	2006–2017	2006–2017	2006–2017	2006–2017
Observations	2,096	3,144	3,144	3,144	3,144
Within R-squared	0.0089	0.0182	0.0002	0.0237	0.0159

Notes: Two-way fixed effects OLS estimates from a difference-in-differences specification, including census block group and year fixed effects. Standard errors clustered at the census block group level in parentheses. Sample period varies by outcome due to differences in data availability. Pre-treatment mean is calculated among treated block groups in the pre-period. Gun crime incidents are incidents involving a firearm and include homicide, aggravated assault with a firearm, robbery with a firearm, and weapon violations. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Difference-in-Differences Results: Part I Crime Incidents

	(1) Part 1 Crimes	(2) Violent Crimes	(3) Property Crimes
Treat $\times$ Post	0.958 (2.675)	-1.859*** (0.488)	2.816 (2.350)
Census Block Group FEs	YES	YES	YES
Year FEs	YES	YES	YES
Pre-Treatment Mean	49.619	13.088	36.531
Sample	2006–2017	2006–2017	2006–2017
Observations	3,144	3,144	3,144
R^2	0.012	0.029	0.014

Notes: Two-way fixed effects OLS estimates from a difference-in-differences specification, including census block group and year fixed effects. Standard errors clustered at the census block group level in parentheses. Pre-treatment mean is calculated among treated block groups in the pre-period. Part I crimes include serious offenses against people and property. Violent crimes include homicide, sexual assault, robbery, and aggravated assault with and without a firearm. Property crimes include burglary, theft from a motor vehicle, larceny, auto theft, and arson. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Difference-in-Differences Results: Part II Crime Incidents

	(1) Part 2 Crimes	(2) Other Persons	(3) Public Disorder	(4) Drug & Alcohol	(5) White Collar	(6) Other
Treat $\times$ Post	-20.123*** (4.756)	-0.849 (0.580)	-3.867*** (1.224)	-2.933*** (0.594)	-1.597 (1.170)	-10.640*** (2.749)
Census Block Group FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES
Pre-Treatment Mean	88.812	17.074	20.119	9.185	6.626	32.393
Sample	2006–2017	2006–2017	2006–2017	2006–2017	2006–2017	2006–2017
Observations	3,144	3,144	3,144	3,144	3,144	3,144
R^2	0.018	0.004	0.030	0.082	0.014	0.020

Notes: Two-way fixed effects OLS estimates from a difference-in-differences specification, including census block group and year fixed effects. Standard errors clustered at the census block group level in parentheses. Pre-treatment mean is calculated among treated block groups in the pre-period. Part II crimes are less serious, generally nonviolent offenses. Other crimes against persons includes assaults with no resulting serious injury, sex offenses, and offenses against the family and children. Public disorder includes criminal mischief (vandalism), prostitution and commercialized vice, drunkenness, disorderly conduct, vagrancy, and curfew and loitering violations. Drug and alcohol offenses include drug abuse violations and liquor law offenses. White collar crimes include forgery, counterfeiting, fraud, and embezzlement. All other crimes include weapons offenses, stolen property offenses, and all other violations of state or local laws not otherwise specifically identified. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Difference-in-Differences Results: Rental Prices and Housing Valuations

	<i>Outcome Variable</i>					
	Rent			Housing Value		
	Low Qrt.	Median	High Qrt.	Low Qrt.	Median	High Qrt.
Treat $\times$ Post	41.600* (23.153)	64.305** (27.565)	83.327** (37.336)	11,480 (6,582)	17,384* (10,488)	12,041 (16,206)
Census Block Group FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES
Pre-Treatment Mean	684.650	869.952	1,091.081	172,463	248,142	337,526
Sample	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019	2009–2019
Observations	2,516	2,567	2,583	2,779	2,813	2,794
Within R-squared	0.0034	0.0055	0.0050	0.0024	0.0023	0.0006

Notes: Two-way fixed effects OLS estimates from a difference-in-differences specification, including census block group and year fixed effects. Standard errors clustered at the census block group level in parentheses. Pre-treatment mean is calculated among treated block groups in the pre-period. Rental price and housing valuation data originate from American Community Survey 5-year estimates. Rent represents the household's monthly contract rent payment; housing valuation represents the household's self-reported estimate of how much their house and lot would sell for if it were for sale. Low Qrt., Median, and High Qrt. report the 25th percentile, 50th percentile, and 75th percentile of the respective outcome. Rents and housing valuations are reported in 2019 dollars. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.4 Comparison

Table 10: Robustness: Effects of Focused Deterrence on Crime Across Specifications

	<i>Outcome Variable</i>				
	Shooting Victims	Total Crime	Part I Crime	Part II Crime	Gun Crime
<i>Panel A: PSM – Trends (Main Specification)</i>					
Treat $\times$ Post	−0.373*** (0.091)	−19.165*** (4.489)	0.958 (2.679)	−20.123*** (4.762)	−1.298*** (0.247)
% Effect	−41.4%	−13.8%	1.9%	−22.7%	−24.2%
<i>Panel B: PSM – Levels Only</i>					
Treat $\times$ Post	−0.297*** (0.096)	−29.460*** (6.501)	−0.972 (1.502)	−28.488*** (6.462)	−1.349*** (0.250)
% Effect	−32.4%	−20.8%	−1.9%	−31.4%	−24.5%
<i>Panel C: PSM – Levels and Trends</i>					
Treat $\times$ Post	−0.253*** (0.094)	−9.690** (4.449)	−1.553 (1.211)	−8.137** (3.738)	−0.590* (0.320)
% Effect	−29.1%	−7.0%	−3.1%	−9.1%	−11.1%
<i>Panel D: AIPW Weighting</i>					
Treat $\times$ Post	−0.282*** (0.082)	−15.484*** (5.277)	1.250 (1.998)	−16.734*** (5.087)	−0.853*** (0.256)
% Effect	−30.4%	−10.6%	2.4%	−17.7%	−15.4%
Census Block Group FEs	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES

Notes: Each panel reports TWFE OLS estimates from a distinct comparison group construction. Panel A is the preferred specification, matching on block group demographics and pre-period crime trends (2006–2012). Panel B matches on demographics and average pre-period crime levels (2009–2012). Panel C matches on demographics, crime levels, and crime trends jointly. Panel D applies augmented inverse probability weights derived from the main propensity score model to the full Philadelphia sample. Percent effects computed as the coefficient divided by the panel-specific pre-treatment mean. Standard errors clustered at the block group level in parentheses. Sample periods: 2010–2017 (shooting victims) and 2006–2017 (all other outcomes). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11: Robustness: Effects of Focused Deterrence on Rents and Housing Values Across Specifications

	<i>Outcome Variable</i>					
	Rent			Housing Value		
	Low Qrt.	Median	High Qrt.	Low Qrt.	Median	High Qrt.
<i>Panel A: PSM – Trends (Main Specification)</i>						
Treat × Post	41.600* (23.153)	64.305** (27.565)	83.327** (37.336)	11,480 (6,582)	17,384* (10,488)	12,041 (16,206)
% Effect	6.1%	7.4%	7.6%	6.7%	7.0%	3.6%
<i>Panel B: PSM – Levels Only</i>						
Treat × Post	67.689*** (23.202)	106.411*** (26.389)	127.947*** (35.641)	20,725*** (6,627)	19,346** (9,431)	16,005 (14,110)
% Effect	9.8%	12.1%	11.7%	12.1%	8.0%	5.0%
<i>Panel C: PSM – Levels and Trends</i>						
Treat × Post	18.008 (23.479)	42.196 (25.674)	86.350** (35.742)	7,433 (6,379)	8,769 (8,549)	2,467 (14,797)
% Effect	2.7%	5.0%	8.1%	4.5%	3.7%	0.8%
<i>Panel D: AIPW Weighting</i>						
Treat × Post	43.753** (18.658)	74.125*** (21.742)	109.359*** (30.068)	13,789** (5,631)	14,175* (7,541)	7,203 (11,519)
% Effect	6.4%	8.4%	9.9%	7.9%	5.8%	2.2%
Census Block Group FEs	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES
Sample	2009–2019					

Notes: Each panel reports TWFE OLS estimates from a distinct comparison group construction. Panel A is the preferred specification, matching on block group demographics and pre-period crime trends (2006–2012). Panel B matches on demographics and average pre-period crime levels (2009–2012). Panel C matches on demographics, crime levels, and crime trends jointly. Panel D applies augmented inverse probability weights derived from the main propensity score model to the full Philadelphia sample. Percent effects computed as the coefficient divided by the panel-specific pre-treatment mean. Standard errors clustered at the block group level in parentheses. Rents and housing valuations measured in dollars. *** p<0.01, ** p<0.05, * p<0.10.